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United States Patent	12413643
Kind Code	B2
Date of Patent	September 09, 2025
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Analyzing augmented reality content usage data

Abstract

In one or more implementations, user interfaces may be generated that include user interface elements that enable the tracking or monitoring of interactions with augmented reality content by users of a client application. Usage metrics for augmented reality content may be collected and analyzed. In addition, information about client application users that interact with augmented reality content may be generated. In one or more examples, characteristics of users of the client application that interact with one or more augmented reality content items may be analyzed to determine a user characterization for one or more augmented reality content items. The usage metrics and user characterizations for augmented reality content items may be presented via a dashboard that is accessible to creators of augmented reality content.

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Appl. No.:	18/665936
Filed:	May 16, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240305691 A1	Sep. 12, 2024

Related U.S. Application Data

continuation parent-doc US 18310367 20230501 US 12015671 child-doc US 18665936
continuation parent-doc US 17362623 20210629 US 11641403 child-doc US 18310367
us-provisional-application US 62705468 20200629

Publication Classification

Int. Cl.: **G06F3/0482** (20130101); **G06F3/01** (20060101); **G06T19/00** (20110101); **H04L67/131** (20220101); **H04L67/306** (20220101); **H04L67/50** (20220101)

U.S. Cl.:

CPC **H04L67/131** (20220501); **G06F3/011** (20130101); **G06F3/0482** (20130101); **G06T19/006** (20130101); **H04L67/306** (20130101); **H04L67/535** (20220501); **G06T2200/24** (20130101); **G06T2219/024** (20130101)

Field of Classification Search

CPC: H04L (67/131); G06F (3/11); G06F (3/0482)

USPC: 709/220; 709/224-228

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 18/310,367, filed May 1, 2023, which application is a continuation of U.S. patent application Ser. No. 17/362,623, filed Jun. 29, 2021, now issued as U.S. Pat. No. 11,641,403, which patent application claims the benefit of U.S. Provisional Patent Application No. 62/705,468, filed Jun. 29, 2020, entitled “ANALYZING AUGMENTED REALITY CONTENT USAGE DATA”, which are incorporated by reference herein in their entirety.

BACKGROUND

(1) Applications executed by client devices may be used to generate content. For example, client applications may be used to generate messaging content, image content, video content, audio content, media overlays, documents, creative works, combinations thereof, and the like. In various situations, user content may be modified by augmented reality content.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some implementations are illustrated by way of example, and not limitation.
- (2) FIG. 1 is a diagrammatic representation of an architecture for exchanging data (e.g., messages and associated content) over a network, according to one or more example implementations.
- (3) FIG. 2 is a diagrammatic representation of a system, in accordance with some examples, that may have both client-side and server-side functionality.
- (4) FIG. 3 is a schematic diagram illustrating data that may be stored in a database of a server system, according to one or more example implementations.
- (5) FIG. 4 is a schematic diagram illustrating an example framework for content that may be accessible via a client application, in accordance with one or more example implementations.
- (6) FIG. 5 is a diagrammatic representation illustrating an architecture to analyze augmented reality content usage data, in accordance with one or more example implementations.
- (7) FIG. 6 is a flowchart illustrating example operations of a process to determine a characterization of client application users that interact with one or more augmented reality content items, according to one or more example implementations.
- (8) FIG. 7 is an illustration of a user interface that includes features that may be incorporated into augmented reality content, according to one or more example implementations.
- (9) FIG. 8 is an illustration of a user interface that includes user content to which an augmented reality content item may be applied, according to one or more example implementations.
- (10) FIG. 9 is an illustration of a user interface that includes augmented reality content that is applied to user content with a number of features of the augmented reality content being associated with a selectable user interface element, according to one or more example implementations.
- (11) FIG. 10 is a representation of an environment that includes a client device that displays a user interface including information about one or more augmented reality content items of an AR content creator, according to one or more example implementations.
- (12) FIG. 11 is a representation of an environment that includes a client device that displays an additional user interface including information about one or more augmented reality content items of an AR content creator, according to one or more example implementations.
- (13) FIG. 12 is a representation of an environment that includes a client device that displays a user interface including information that characterizes users of a client application that interact with augmented reality content items, according to one or more example implementations.
- (14) FIG. 13 is a block diagram illustrating components of a machine, in the form of a computer system, that may read and execute instructions from one or more machine-readable media to perform any one or more methodologies described herein, in accordance with one or more example implementations.
- (15) FIG. 14 is block diagram illustrating a representative software architecture that may be used in conjunction with one or more hardware architectures described herein, in accordance with one or more example implementations.

DETAILED DESCRIPTION

- (16) Individuals and organizations may create augmented reality content that may be consumed by users of a client application that may execute the augmented reality content. For example, augmented reality content may be created that modifies an appearance of one or more objects included in user content, where the user content may include at least one of image content, video

content, or audio content captured via one or more input devices of a client device. To illustrate, augmented reality content may modify an appearance of one or more individuals included in user content. In one or more examples, the augmented reality content may modify user content by adding at least one of image content or video content to user content that modifies an appearance of an object included in the user content. In various examples, the augmented reality content may overlay at least a portion of an object included in the user content. Additionally, the augmented reality content may modify pixel data of image content or video content to change an appearance of at least one object included in the user content.

(17) In existing systems and methods, the amount of information that augmented reality content creators may obtain about the use of their augmented reality content is limited. In one or more examples, augmented reality content creators may obtain information indicating an amount of time that users of a client application have viewed the augmented reality content or a number of times that the augmented reality content has been shared with additional users of the client application. In addition, augmented reality content creators are, at best, able to obtain limited information about the characteristics of users of the client application that are consuming the augmented reality content generated by these creators. Existing systems and methods generate augmented reality content that lacks the technical functionality to track the usage of augmented reality content in ways that provide more robust information to augmented reality content creators about the use of the augmented reality content they have created. In particular, during the creation of augmented reality content, existing systems and methods provide limited granularity with respect to the interactions that users have with augmented reality content. Further, the technical infrastructure used by existing systems and methods limits the availability and the amount of information about characteristics of users that interact with augmented reality content.

(18) The systems, methods, techniques, instruction sequences, and computing machine program products described herein are directed to collecting and analyzing augmented reality content usage data. The implementations described herein may be used to obtain additional information related to the use of augmented reality content that is not available with existing systems and methods. For example, implementations described herein may collect profile information of users of a client application that interact with augmented reality content. Implementations described herein may also analyze the information collected about the client application users. In this way, the systems and methods described herein may provide detailed information to augmented reality content creators about the users that consume the augmented reality content generated by these creators. In one or more scenarios, augmented reality content creators may tailor their augmented reality content to an audience of additional client application users based on the characteristics of client application users that have previously interacted with their augmented reality content. Additionally, augmented reality content creators may implement campaigns to bring awareness of their augmented reality content to additional client application users that share characteristics with client application users that have previously interacted with the augmented reality content generated by these creators.

(19) In various examples, the systems and methods described herein may enable creators of augmented reality content to tie a number of specific user interface elements to features of augmented reality content items. For example, creators of augmented reality content may select features of their augmented reality content items for which to track user interactions with the features. To illustrate, an augmented reality content item may include a number of features that client application users may interact with and a creator of the augmented reality content item may indicate one or more of the features to associate with a selectable user interface element. In these situations, user input indicating that at least one of the selectable user interface elements has been chosen may be collected and analyzed. In this way, additional interactions by client application users with augmented reality content may be tracked that are not able to be captured by existing systems and methods and a greater degree of granularity with respect to interactions with features

of augmented reality content items may be tracked in relation to existing systems and methods. In one or more examples, metrics indicating the amount of user interaction with respect to an augmented reality content item or the amount of user interaction with features of the augmented reality content item may be determined based on the data obtained from tracking the selection of user interface elements that correspond to the augmented reality content item.

(20) The systems and methods described herein may also determine profile information of client application users that interact with augmented reality content. In various examples, implementations described herein may identify interactions by client application users with augmented reality content and store profile information of the client applications users in association with the augmented reality content. The systems and methods described herein may then analyze the user profile information to determine a characterization of client application users that interact with the augmented reality content. The characterization of client application users may indicate characteristics of client application users that interact with the augmented reality content. In one or more examples, the profile information of client application users that interact with the augmented reality content may be analyzed to determine a target audience for the augmented reality content.

(21) Accordingly, the systems, methods, techniques, instruction sequences, and computing machine program products described herein provide various implementations to collect more information about the use of augmented reality content than existing systems and methods by providing an underlying technical architecture that enables the tracking of interactions by client application users with various features of augmented reality content items. The systems and methods described herein also provide additional insights in relation to existing systems and methods for augmented reality content creators with regard to the client application users that interact with the augmented reality content items generated by these creators. In this way, the augmented reality content creators may identify client application users that are more likely to interact with their augmented reality content than existing systems and methods. In situations where the augmented reality content is related to one or more products, the augmented reality content creators are more likely to identify client application users that will purchase their products than existing systems and methods.

(22) FIG. 1 is a diagrammatic representation of an architecture **100** for exchanging data (e.g., messages and associated content) over a network. The architecture **100** may include multiple client devices **102**. The client devices **102** may individually comprise, but are not limited to, a mobile phone, a desktop computer, a laptop computing device, a portable digital assistant (PDA), smart phone, tablet computing device, ultrabook, netbook, multi-processor system, microprocessor-based or programmable consumer electronic system, game console, set-top box, computer in a vehicle, a wearable device, one or more combinations thereof, or any other communication device that a user may utilize to access one or more components included in the architecture **100**.

(23) Each client device **102** may host a number of applications, including a client application **104** and one or more third-party applications **106**. A user may use the client application **104** to create content, such as video, images (e.g., photographs), audio, and media overlays. In one or more illustrative examples, the client application **104** may include a social networking functionality that enables users to create and exchange content. In various examples, the client application **104** may include messaging functionality that may be used to send messages between instances of the client application **104** executed by various client devices **102**. The messages created using the client application **104** may include video, one or more images, audio, media overlays, text, content produced using one or more creative tools, annotations, and the like. In one or more implementations, the client application **104** may be used to view and generate interactive messages, view locations of other users of the client application **104** on a map, chat with other users of the client application **104**, and so forth.

(24) One or more users may be a person, a machine, or other means of interacting with a client device, such as the client device **102**. In example implementations, the user may not be part of the

architecture **100** but may interact with one or more components of the architecture **100** via a client device **102** or other means. In various examples, users may provide input (e.g., touch screen input or alphanumeric input) to a client device **102** and the input may be communicated to other entities in the architecture **100**. In this instance, the other entities in the architecture **100**, responsive to the user input, may communicate information to a client device **102** to be presented to the users. In this way, users may interact with the various entities in the architecture **100** using the client device **102**.

(25) Each instance of the client application **104** is able to communicate and exchange data with at least one of another instance of the client application **104**, one or more third-party applications **106**, or a server system **108**. The data exchanged between instances of the client applications **104**, between the third-party applications **106**, and between instances of the client application **104** and the server system **108** includes functions (e.g., commands to invoke functions) and payload data (e.g., text, audio, image, video, or other multimedia data). Data exchanged between instances of the client applications **104**, between the third-party applications **106**, and between at least one instance of the client application **104** and at least one third-party application **106** may be exchanged directly from an instance of an application executed by a client device **102** and an instance of an application executed by an additional client device **102**. Further, data exchanged between the client applications **104**, between the third-party applications **106**, and between at least one client application **104** and at least one third-party application **106** may be communicated indirectly (e.g., via one or more intermediate servers) from an instance of an application executed by a client device **102** to another instance of an application executed by an additional client device **102**. In one or more illustrative examples, the one or more intermediate servers used in indirect communications between applications may be included in the server system **108**.

(26) The third-party application(s) **106** may be separate and distinct from the client application **104**. The third-party application(s) **106** may be downloaded and installed by the client device **102** separately from the client application **104**. In various implementations, the third-party application(s) **106** may be downloaded and installed by the client device **102** before or after the client application **104** is downloaded and installed. The third-party application(s) **106** may be an application that is provided by an entity or organization that is different from the entity or organization that provides the client application **104**. The third-party application(s) **106** may be accessed by the client device **102** using separate login credentials than the client application **104**. Namely, the third-party application(s) **106** may maintain a first user account and the client application **104** may maintain a second user account. In one or more implementations, the third-party application(s) **106** may be accessed by the client device **102** to perform various activities and interactions, such as listening to music, videos, track exercises, view graphical elements (e.g., stickers), communicate with other users, and so forth. As an example, the third-party application(s) **106** may include a social networking application, a dating application, a ride or car sharing application, a shopping application, a trading application, a gaming application, an imaging application, a music application, a video browsing application, an exercise tracking application, a health monitoring application, a graphical element or sticker browsing application, or any other suitable application.

(27) The server system **108** provides server-side functionality via one or more networks **110** to the client application **104**. The server system **108** may be a cloud computing environment, according to some example implementations. For example, the server system **108**, and one or more servers associated with the server system **108**, may be associated with a cloud-based application, in one illustrative example. In one or more implementations, the client device **102** and the server system **108** may be coupled via the one or more networks **110**.

(28) The server system **108** supports various services and operations that are provided to the client application **104**. Such operations include transmitting data to, receiving data from, and processing data generated by the client application **104**. This data may include message content, media content, client device information, geolocation information, media annotation and overlays,

message content persistence conditions, social network information, and live event information, as examples. Data exchanges within the architecture **100** are invoked and controlled through functions available via user interfaces (UIs) of the client application **104**.

(29) While certain functions of the architecture **100** are described herein as being performed by either a client application **104** or by the server system **108**, the location of functionality either within the client application **104** or the server system **108** is a design choice. For example, it may be technically preferable to initially deploy certain technology and functionality within the server system **108**, but to later migrate this technology and functionality to the client application **104** where a client device **102** has a sufficient processing capacity.

(30) The server system **108** includes an Application Programming Interface (API) server **112** that is coupled to, and provides a programmatic interface to, an application server **114**. The application server **114** is communicatively coupled to a database server **116** that facilitates access to one or more databases **118**. The one or more databases **118** may store data associated with information processed by the application server **114**. The one or more databases **118** may be storage devices that store information such as untreated media content, original media content from users (e.g., high-quality media content), processed media content (e.g., media content that is formatted for sharing with client devices **102** and viewing on client devices **102**), context data related to a media content item, context data related to a user device (e.g., a computing or client device **102**), media overlays, media overlay smart widgets or smart elements, user data, user device information, media content (e.g., video and images), media content data (e.g., data associated with video and images), computing device context data, serialized data, session data items, user device location data, mapping information, interactive message usage data, interactive message metrics data, and so forth. The one or more databases **118** may further store information related to third-party servers, client devices **102**, client applications **104**, users, third-party applications **106**, and so forth.

(31) The API server **112** receives and transmits data (e.g., commands and message payloads) between client devices **102** and the application server **114**. Specifically, the Application Program Interface (API) server **112** provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the client application **104** in order to invoke functionality of the application server **114**. The Application Program Interface (API) server **112** exposes various functions supported by the application server **114**, including account registration, login functionality, the sending of messages, via the application server **114**, from one instance of the client application **104** to another instance of the client application **104**, the sending of media files (e.g., images, audio, video) from a client application **104** to the application server **114**, and for possible access by another client application **104**, the setting of a collection of media content (e.g., a gallery, story, message collection, or media collection), the retrieval of a list of friends of a user of a client device **102**, the retrieval of such collections, the retrieval of messages and content, the adding and deletion of friends to a social graph, the location of friends within a social graph, and opening an application event (e.g., relating to the client application **104**).

(32) The server system **108** may also include a web server **120**. The web server **120** is coupled to the application servers **114**, and provides web-based interfaces to the application servers **114**. To this end, the web server **120** processes incoming network requests over the Hypertext Transfer Protocol (HTTP) and several other related protocols.

(33) The application server **114** hosts a number of applications and subsystems, including a messaging application system **122**, a media content processing system **124**, a social network system **126**, an augmented reality (AR) content creation system **128**, an AR content usage system **130**, and a content management system **132**. The messaging application system **122** implements a number of message processing technologies and functions, particularly related to the aggregation and other processing of content (e.g., textual and multimedia content) included in messages received from multiple instances of the client application **104**. For example, the messaging application system **122** may deliver messages using electronic mail (email), instant messaging (IM), Short Message

Service (SMS), text, facsimile, or voice (e.g., Voice over IP (VoIP)) messages via wired networks (e.g., the Internet), plain old telephone service (POTS), or wireless networks (e.g., mobile, cellular, WIFI, Long Term Evolution (LTE), or Bluetooth). The messaging application system **122** may aggregate text and media content from multiple sources into collections of content. These collections are then made available, by the messaging application system **122**, to the client application **104**. Other processor- and memory-intensive processing of data may also be performed server-side by the messaging application system **122**, in view of the hardware requirements for such processing.

(34) The media content processing system **124** is dedicated to performing various media content processing operations, typically with respect to images, audio, or video received within the payload of a message or other content item at the messaging application system **122**. The media content processing system **124** may access one or more data storages (e.g., the database(s) **118**) to retrieve stored data to use in processing media content and to store results of processed media content.

(35) The social network system **126** supports various social networking functions and services, and makes these functions and services available to the messaging application system **122**. To this end, the social network system **126** maintains and accesses an entity graph within the database(s) **118**. Examples of functions and services supported by the social network system **126** include the identification of other users of the client application **104** with which a particular user has relationships or is “following”, and also the identification of other entities and interests of a particular user. The social network system **126** may access location information associated with each of the user's friends or other social network connections to determine where they live or are currently located geographically. In addition, the social network system **126** may maintain a location profile for each of the user's friends indicating the geographical location where the user's friends live.

(36) The AR content creation system **128** may be used to generate AR content **134** that includes a number of AR content items **136**. Individual AR content items **136** may include computer-readable instructions that are executable to modify user content captured by the client device **102**. In various examples, the AR content creation system **128** may provide one or more user interfaces that may be used to create the AR content **134**. For example, the AR content creation system **128** may provide one or more user interfaces to import content that may be used to create one or more of the AR content items **136**. To illustrate, the AR content creation system **128** may provide one or more user interfaces to import at least one of image content, video content, or audio content that may be used to generate an AR content item **136**. The AR content creation system **128** may also provide one or more user interfaces with one or more creative tools that may be used to create one or more of the AR content items **136**. In one or more illustrative examples, the AR content creation system **128** may implement at least one of a drawing tool, a writing tool, a sketching tool, or a coloring tool that may be used to generate the AR content **134**.

(37) The AR content creation system **128** may also enable a creator of an AR content item **136** to provide an indication that interactions by users of the client application **104** with the AR content item **136** are to be monitored or tracked. In one or more examples, the AR content creation system may enable a creator of the AR content item **136** to identify one or more features of the AR content item **136** to be tracked. The features of AR content items **136** that are being tracked may be included in tracked features **138**. In one or more illustrative examples, the AR content creation system **128** may provide one or more user interfaces that enable the selection of one or more features of the AR content items **136** that are to be tracked. In various examples, a user interface tool, such as a cursor or pointer, may be used to indicate a feature of an AR content item **136** or a region of an AR content item **136** for which user interactions are to be tracked. Tracking the use of AR content items **136** may comprise at least one of identifying or counting interactions with the AR content items **136** by users of the client application **104**. Interactions with an AR content item **136** may include at least one of applying an AR content item **136** to user content, sharing an AR

content item **136** with another user of the client application **104**, generating a message that includes the AR content item **136** or a link corresponding to the AR content item **136**, activating a feature of the AR content item **136**, selecting a user interface element that corresponds to a feature of the AR content item **136**, unlocking the AR content item **136**, purchasing the AR content item **136**, or selecting an icon that corresponds to the AR content item **136**.

(38) In one or more implementations, the AR content creation system **128** may provide one or more templates that indicate features of AR content **134** that may be tracked. The AR content creation system **128** may provide one or more templates for specified classifications of AR content **134**. For example, the AR content creation system **128** may provide a template for AR content items **136** that include faces, where the template indicates that one or more facial features included in the AR content item **136** may be tracked. In one or more additional examples, the AR content creation system **128** may provide a template for AR content items **136** that include products that may be purchased via the client application **104**. In these scenarios, the AR content creation system **128** may provide a template that indicates one or more features of the product, such as features that enable magnification of a view of the product, features that enable rotation of a view of the product, or features that enable the product to be applied to one or more objects. In one or more scenarios where the AR content item **136** is related to a beauty product, the AR content creation system **128** may provide a template that enables the beauty product to be applied to one or more facial features of an individual included in a user content item.

(39) In various examples, the tracked features **138** may be associated with user interface elements that a user of the client application **104** may interact with. In one or more examples, the tracked features **138** may correspond to a portion of a user interface that may detect input by a user of the client application **104**. The input may be detected by using an input device, such as at least one of a cursor, stylus, finger, another implement, or a speaker of a client device **102**. The input may be directed to selection of a user interface element, moving the user interface element, or another method of indicating an interaction with a tracked feature **138**. In this way, input detected by a user interface element corresponding to a tracked feature **138** of an AR content item **136** may be collected, stored, such as in the database(s) **118**, and analyzed.

(40) The AR content usage system **130** may determine an amount of usage of AR content **134**. The amount of usage of the AR content **134** may be determined based on an amount of interaction with one or more tracked features **138** of each AR content item **136** included in the AR content **134**. Each interaction by a user of the client application **104** with a tracked feature **138** may be counted by the AR content usage system **130** and stored in the database(s) **118**. In various examples, the database(s) **118** may include one or more data structures for individual AR content items **136**, and the one or more data structures for each AR content item **136** may indicate a number of interactions by users of the client application **104** with one or more of the tracked features **138**. In one or more examples, an AR content item **136** may have multiple tracked features **138**. In these scenarios, a data structure corresponding to the AR content item **136** may indicate at least one of interactions related to each of the tracked features **138** of the AR content item **136** or interactions related to a combination of the tracked features **138** of the AR content item **136**.

(41) In one or more examples, the AR content usage system **130** may also track one or more metrics related to the usage of one or more tracked features **138** of one or more AR content items **136** included in the AR content **134**. For example, the AR content usage system **130** may determine a number of times that the AR content item **136** was shared with another user of the client application **104** and a number of times that the AR content item **136** was included in a message sent to another user of the client application **104**. Additionally, the AR content usage system **130** may determine a number of times that the AR content item **136** was applied to user content and a number of times that a tracked feature **138** of the AR content item **136** was selected by a user of the client application **104**. In additional examples, the AR content usage system **130** may collectively track the usage of a number of the tracked features **138** of one or more AR content items **136**

included in the AR content **134**. To illustrate, the AR content usage system **130** may determine a total number of interactions related to the AR content item **136**. In these situations, the AR content usage system **130** may determine a sum that includes a number of times that the AR content item **136** has been shared, a number of times that the AR content item **136** has been related to a message, a number of times that the AR content item **136** has been applied to user content, and a number of times that tracked features **138** of the AR content item **136** have been interacted with by users of the client application **104**.

(42) In various examples, the AR content usage data **140** may be associated with user profile data **144**. For example, the AR content usage system **130** may identify users of the client application **104** that interact with the AR content **134**. In one or more examples, identifiers of users of the client application **104** may be determined by the AR content usage system **130**. Based on identifying users of the client application **104** that interacted with one or more AR content items **136**, the AR content usage system **130** may determine characteristics of the users. In one or more examples, the AR content usage system **130** may determine demographic information related to users that interact with AR content items **136**, such as age, gender, occupation, education, one or more combinations thereof, and the like. In additional examples, the AR content usage system **130** may determine location information of users of the client application **104** that interact with AR content items **136**. In various examples, the location information may be determined based on at least one of user input or geographic positioning system (GPS) data.

(43) In further examples, that AR content usage system **130** may determine additional information about users of the client application **104** that correspond to interactions by the users of client application **104** with content consumed via the client application **104**. For example, the AR content usage system **130** may determine usage metrics for one or more features of the client application **104** in relation to users of the client application **104** that interact with AR content **134**. To illustrate, the AR content usage system **130** may determine an amount of usage of one or more social networking features of the client application **104** by users of the client application **104** that interact with one or more AR content items **136**. The AR content usage system **130** may also determine an amount of usage of one or more messaging features of the client application **104** by users of the client application **104** that interact with one or more AR content items **136**. In addition, the AR content usage system **130** may determine an amount of usage of one or more search features of the client application **104** by users of the client application **104** that interact with one or more AR content items **136**. Further, the AR content usage system **130** may determine sources of content that are accessed by users of the client application **104** that interact with one or more AR content items **136**. The sources of content may include one or more additional users of the client application **104**, one or more media organizations, one or more news organizations, one or more fashion organizations, one or more retailers, one or more manufacturers, one or more governmental organizations, one or more authors, one or more blogs, one or more websites, one or more periodicals, one or more combinations thereof, and so forth. In various examples, the AR content usage system **130** may also determine types of AR content **134** that are interacted with by users of the client application **104**. In one or more illustrative examples, the AR content usage system **130** may determine interactions with AR content items **136** that comprise overlays of at least one of image content or video content by users of the client application **104**. Additionally, the AR content usage system **130** may determine interactions with AR content items **136** that include animations created by users of the client application **104**.

(44) In one or more implementations, the AR content usage system **130** may determine characterizations of users of the client application **104** that interact with one or more AR content items **136**. In one or more examples, the AR content usage system **130** may determine one or more characteristics of users of the client application **104** that interact with an AR content item **136**. In various examples, the AR content usage system **130** may determine a characterization of users of the client application **104** that interact with the AR content item **136** by determining one or more

characteristics that are associated with at least a threshold number of the users of the client application **104** that interact with the AR content item **136**. For example, the AR content usage system **130** may determine one or more demographic characteristics that are common to at least a threshold number of users of the client application **104** that interact with the AR content item **136**. The AR content usage system **130** may also determine one or more location characteristics that are common to at least a threshold number of users of the client application **104** that interact with the AR content item **136**. In additional examples, the AR content usage system **130** may determine one or more AR content usage characteristics that are common to at least a threshold number of users of the client application **104** that interact with the AR content item **136**.

(45) Based on characteristics of users of the client application **104** that interact with the AR content item **136**, the AR content usage system **130** may determine a target audience for the AR content item **136**. The target audience for the AR content item **136** may correspond to users of the client application **104** that have at least a threshold likelihood of interacting with the AR content item **136**. In this way, the characterization of users of the client application **104** that may interact with the AR content item **136** may provide insight to one or more creators of the AR content item **136**. Thus, the one or more creators of the AR content item **136** may identify users of the client application **104** that have not previously interacted with the AR content item **136**, but have at least a threshold likelihood of interacting with the AR content item **136**. In these scenarios, the one or more creators of the AR content item **136** may cause the server system **108** to bring awareness to the target audience of the AR content item **136**. The server system **108** may bring awareness of the AR content item **136** to the target audience via at least one of advertisements or recommendations related to the AR content item **136**. Further, the server system **108** may bring awareness of the AR content item **136** to the target audience by increasing the weighting of the AR content item **136** in search results for AR content **134** submitted by users of the client application **104** included in the target audience.

(46) The content management system **132** may generate user interface data that corresponds to user interfaces that include information about usage metrics with respect to one or more AR content items **136**. The content management system **132** may also generate user interface data that includes information indicating characteristics of users of the client application **104** that interact with one or more AR content items **136**. In various examples, creators of AR content **134** may send requests to the server system **108** for information about users of the client application **104** that interact with one or more AR content items **136** generated by the creators. The requests may be directed to metrics related to specified characteristics of users of the client application **104** that interact with one or more AR content items **136** generated by an augmented reality content creator. For example, an augmented reality content creator may send a request to the server system **108** to obtain at least one of age, location, gender, occupation, or education of users of the client application **104** that interact with one or more AR content items **136** of the augmented reality content creator. In an additional example, the server system **108** may receive a request from an augmented reality content creator to obtain at least one of a frequency of interaction with AR content **134** generally or a frequency of interaction of a type of AR content **134** interacted with by users of the client application **104** that also interact with one or more AR content items **136** of the augmented reality content creator. In various examples, the content management system **132** may provide one or more templates that may be used by augmented reality content creators to request information about at least one of interactions with AR content items **136** of the creators or characteristics of users of the client application **104** that interact with AR content items **136** of the creator.

(47) In response to a request from an augmented reality content creator for information about one or more of the augmented reality content items produced by the creator, the content management system **132** may operate in conjunction with the AR content usage system **130** to determine at least one of augmented reality content usage metrics or characteristics of users of the client application **104** related to the AR content **134** of the creator. The content management system **132** may then

generate user interface data that indicates the information satisfying the request. In various examples, the content management system **132** may generate user interface data that corresponds to a dashboard that is accessible to augmented reality content creators, where the dashboard includes information corresponding to an information request received by the server system **108** from the creator. In one or more examples, the dashboard may include one or more standardized features that include information about the use of AR content **134** that are provided to each augmented reality content creator requesting augmented reality content information from the server system **108**. In additional examples, the dashboard may include one or more customized features that include information about the use of AR content **134** based on specific requests of augmented reality content creators that are received by the server system **108**.

(48) FIG. 2 is a block diagram illustrating further details regarding the server system **108**, according to some examples. Specifically, the server system **108** is shown to comprise the client application **104** and the application servers **114**. The server system **108** embodies a number of subsystems, which are supported on the client-side by the client application **104** and on the sever-side by the application servers **114**. These subsystems include, for example, an ephemeral timer system **202**, a collection management system **204**, an augmentation system **206**, a map system **208**, a game system **210**, the AR content creation system **128**, the AR content usage system **130**, and the content management system **132**.

(49) The ephemeral timer system **202** is responsible for enforcing the temporary or time-limited access to content by the client application **104** and the messaging application system **122**. The ephemeral timer system **202** incorporates a number of timers that, based on duration and display parameters associated with a message, or collection of messages (e.g., a story), selectively enable access (e.g., for presentation and display) to messages and associated content via the client application **104**. Further details regarding the operation of the ephemeral timer system **202** are provided below.

(50) The collection management system **204** is responsible for managing sets or collections of media (e.g., collections of text, image video, and audio data). A collection of content (e.g., messages, including images, video, text, and audio) may be organized into an “event gallery” or an “event story.” Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a “story” for the duration of that music concert. The collection management system **204** may also be responsible for publishing an icon that provides notification of the existence of a particular collection to the user interface of the client application **104**.

(51) The collection management system **204** furthermore includes a curation interface **212** that allows a collection manager to manage and curate a particular collection of content. For example, the curation interface **212** enables an event organizer to curate a collection of content relating to a specific event (e.g., delete inappropriate content or redundant messages). Additionally, the collection management system **204** employs machine vision (or image recognition technology) and content rules to automatically curate a content collection. In certain examples, compensation may be paid to a user for the inclusion of user-generated content into a collection. In such cases, the collection management system **204** operates to automatically make payments to such users for the use of their content.

(52) The augmentation system **206** provides various functions that enable a user to augment (e.g., annotate or otherwise modify or edit) media content associated with content produced via the client application **104**, such as a message. For example, the augmentation system **206** provides functions related to the generation and publishing of media overlays for content processed by the server system **108**. The augmentation system **206** operatively supplies a media overlay or augmentation (e.g., an image filter) to the client application **104** based on a geolocation of the client device **102**. In another example, the augmentation system **206** operatively supplies a media overlay to the client application **104** based on other information, such as social network information of the user of the

client device **102**. A media overlay may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects. An example of a visual effect includes color overlaying. The audio and visual content or the visual effects can be applied to a media content item (e.g., a photo) at the client device **102**. For example, the media overlay may include text or image that can be overlaid on top of a photograph taken by the client device **102**. In another example, the media overlay includes an identification of a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In another example, the augmentation system **206** uses the geolocation of the client device **102** to identify a media overlay that includes the name of a merchant at the geolocation of the client device **102**. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the database(s) **118** and accessed through the database server(s) **116**.

(53) In some examples, the augmentation system **206** provides a user-based publication platform that enables users to select a geolocation on a map and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The augmentation system **206** generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.

(54) In other examples, the augmentation system **206** provides a merchant-based publication platform that enables merchants to select a particular media overlay associated with a geolocation via a bidding process. For example, the augmentation system **206** associates the media overlay of the highest bidding merchant with a corresponding geolocation for a predefined amount of time.

(55) The map system **208** provides various geographic location functions, and supports the presentation of map-based media content and messages by the client application **104**. For example, the map system **208** enables the display of user icons or avatars (e.g., stored in profile data **308** of FIG. 3) on a map to indicate a current or past location of “friends” of a user, as well as media content (e.g., collections of messages including photographs and videos) generated by such friends, within the context of a map. For example, a message posted by a user to the server system **108** from a specific geographic location may be displayed within the context of a map at that particular location to “friends” of a specific user on a map interface of the client application **104**. A user can furthermore share his or her location and status information (e.g., using an appropriate status avatar) with other users of the server system **108** via the client application **104**, with this location and status information being similarly displayed within the context of a map interface of the client application **104** to selected users.

(56) The game system **210** provides various gaming functions within the context of the client application **104**. The client application **104** provides a game interface providing a list of available games that can be launched by a user within the context of the client application **104**, and played with other users of the server system **108**. The server system **108** further enables a particular user to invite other users to participate in the play of a specific game, by issuing invitations to such other users from the client application **104**. The client application **104** also supports both the voice and text messaging (e.g., chats) within the context of gameplay, provides a leaderboard for the games, and also supports the provision of in-game rewards (e.g., coins and items).

(57) The AR content creation system **128** may generate one or more user interfaces that accept input that may be used to generate augmented reality content. The input obtained via the one or more user interfaces may be used to determine one or more features to associate with augmented reality content. In various examples, the one or more user interfaces generated using data from the AR content creation system **128** may include creative tools that enable users to associate one or more visual effects with augmented reality content. In one or more examples, the one or more user interfaces generated using data obtained from the AR content creation system **128** may associate one or more audio effects with augmented reality content. The augmentation system **206** may operate in conjunction with the AR content creation system **128** to generate augmented reality

content according to the input obtained via the user interfaces generated by the AR content creation system **128**. For example, the augmentation system **206** may operate in conjunction with the AR content creation system **128** to generate computer-readable instructions that correspond to augmented reality content that are executable to produce at least one of visual effects or audio effects in relation to user content. The AR content creation system **128** may also generate user interface data that enables the tracking of interactions with augmented reality content by users of the client application **104**. In one or more examples, the AR content creation system **128** may generate user interface data that enables interactions with one or more features of augmented reality content and that enables the interactions with the one or more features to be recorded.

(58) The AR content usage system **130** may determine an amount of usage of augmented reality content. In one or more implementations, the AR content usage system **130** may determine a number of interactions with the augmented reality content that is executed via the client application **104**. In various examples, the AR content usage system **130** may determine a number of interactions with one or more specified features of augmented reality content that is executed via the client application **104**. In addition, the AR content usage system **130** may determine characteristics of users of the client application **104** that interact with augmented reality content. For example, the AR content usage system **130** may determine characterizations of users of the client application **104** that interact with augmented reality content that indicate at least one of demographic features, location information, usage metrics related to one or more features of the client application **104**, usage metrics related to one or more augmented reality content items, or content consumption information with respect to users of the client application **104** that interact with one or more augmented reality content items.

(59) Further, the content management system **132** may generate data for one or more user interfaces that indicate metrics corresponding to users of the client application **104** that interact with augmented reality content via the client application **104**. In one or more examples, a creator of augmented reality content may request to view one or more metrics related to one or more augmented reality content items generated by the creator. The content management system **132** may, in response to the request, generate user interface data that indicates the requested usage metrics. In various examples, the content management system **132** may operate in conjunction with the AR content usage system **130** to obtain the metrics that correspond to the request and to generate the user interface data that indicates the requested metrics.

(60) FIG. 3 is a schematic diagram illustrating data structures **300** which may be stored in the database(s) **118** of the server system **108**, according to one or more example implementations. While the content of the database(s) **118** is shown to comprise a number of tables, it will be appreciated that the data could be stored in other types of data structures (e.g., as an object-oriented database).

(61) The database **118** includes message data stored within a message table **302**. This message data includes, for any particular one message, at least message sender data, message recipient (or receiver) data, and a payload.

(62) An entity table **304** stores entity data, and is linked (e.g., referentially) to an entity graph **306** and profile data **308**. Entities for which records are maintained within the entity table **304** may include individuals, corporate entities, organizations, objects, places, events, and so forth. Regardless of entity type, any entity regarding which the server system **108** stores data may be a recognized entity. Each entity is provided with a unique identifier, as well as an entity type identifier (not shown).

(63) The entity graph **306** stores information regarding relationships and associations between entities. Such relationships may be social, professional (e.g., work at a common corporation or organization) interested-based or activity-based, merely for example.

(64) The profile data **308** stores multiple types of profile data about a particular entity. The profile data **308** may be selectively used and presented to other users of the architecture **100**, based on

privacy settings specified by a particular entity. Where the entity is an individual, the profile data **308** includes, for example, a username, telephone number, address, settings (e.g., notification and privacy settings), as well as a user-selected avatar representation (or collection of such avatar representations). A particular user may then selectively include one or more of these avatar representations within the content of messages or other data communicated via the architecture **100**, and on map interfaces displayed by client application **104** to other users. The collection of avatar representations may include “status avatars,” which present a graphical representation of a status or activity that the user may select to communicate at a particular time.

(65) Where the entity is a group, the profile data **308** for the group may similarly include one or more avatar representations associated with the group, in addition to the group name, members, and various settings (e.g., notifications) for the relevant group.

(66) The database **118** also stores augmentation data, such as overlays or filters, in an augmentation table **310**. The augmentation data is associated with and applied to videos (for which data is stored in a video table **314**) and images (for which data is stored in an image table **316**).

(67) Filters, in one example, are overlays that are displayed as overlaid on an image or video during presentation to a recipient user. Filters may be of various types, including user-selected filters from a set of filters presented to a sending user by the client application **104** when the sending user is composing a message. Other types of filters include geolocation filters (also known as geo-filters), which may be presented to a sending user based on geographic location. For example, geolocation filters specific to a neighborhood or special location may be presented within a user interface by the client application **104**, based on geolocation information determined by a Global Positioning System (GPS) unit of the client device **102**.

(68) Another type of filter is a data filter, which may be selectively presented to a sending user by the client application **104**, based on other inputs or information gathered by the client device **102** during the message creation process. Examples of data filters include current temperature at a specific location, a current speed at which a sending user is traveling, battery life for a client device **102**, or the current time.

(69) Other augmentation data that may be stored within the image table **316** includes augmented reality content items (e.g., corresponding to applying Lenses or augmented reality experiences). An augmented reality content item may be a real-time special effect and sound that may be added to an image or a video.

(70) As described above, augmentation data includes augmented reality content items, overlays, image transformations, AR images, and similar terms refer to modifications that may be applied to image data (e.g., videos or images). This includes real-time modifications, which modify an image as it is captured using device sensors (e.g., one or multiple cameras) of a client device **102** and then displayed on a screen of the client device **102** with the modifications. This also includes modifications to stored content, such as video clips in a gallery that may be modified. For example, in a client device **102** with access to multiple augmented reality content items, a user can use a single video clip with multiple augmented reality content items to see how the different augmented reality content items will modify the stored clip. For example, multiple augmented reality content items that apply different pseudorandom movement models can be applied to the same content by selecting different augmented reality content items for the content. Similarly, real-time video capture may be used with an illustrated modification to show how video images currently being captured by sensors of a client device **102** would modify the captured data. Such data may simply be displayed on the screen and not stored in memory, or the content captured by the device sensors may be recorded and stored in memory with or without the modifications (or both). In some systems, a preview feature can show how different augmented reality content items will look within different windows in a display at the same time. This can, for example, enable multiple windows with different pseudorandom animations to be viewed on a display at the same time.

(71) Data and various systems using augmented reality content items or other such transform

systems to modify content using this data can thus involve detection of objects (e.g., faces, hands, bodies, cats, dogs, surfaces, objects, etc.), tracking of such objects as they leave, enter, and move around the field of view in video frames, and the modification or transformation of such objects as they are tracked. In various implementations, different methods for achieving such transformations may be used. Some examples may involve generating a three-dimensional mesh model of the object or objects, and using transformations and animated textures of the model within the video to achieve the transformation. In other examples, tracking of points on an object may be used to place an image or texture (which may be two dimensional or three dimensional) at the tracked position. In still further examples, neural network analysis of video frames may be used to place images, models, or textures in content (e.g., images or frames of video). Augmented reality content items thus refer both to the images, models, and textures used to create transformations in content, as well as to additional modeling and analysis information needed to achieve such transformations with object detection, tracking, and placement.

(72) Real-time video processing can be performed with any kind of video data (e.g., video streams, video files, etc.) saved in a memory of a computerized system of any kind. For example, a user can load video files and save them in a memory of a device, or can generate a video stream using sensors of the device. Additionally, any objects can be processed using a computer animation model, such as a human's face and parts of a human body, animals, or non-living things such as chairs, cars, or other objects.

(73) In some examples, when a particular modification is selected along with content to be transformed, elements to be transformed are identified by the computing device, and then detected and tracked if they are present in the frames of the video. The elements of the object are modified according to the request for modification, thus transforming the frames of the video stream. Transformation of frames of a video stream can be performed by different methods for different kinds of transformation. For example, for transformations of frames mostly referring to changing forms of object's elements characteristic points for each element of an object are calculated (e.g., using an Active Shape Model (ASM) or other known methods). Then, a mesh based on the characteristic points is generated for each of the at least one element of the object. This mesh is used in the following stage of tracking the elements of the object in the video stream. In the process of tracking, the mentioned mesh for each element is aligned with a position of each element. Then, additional points are generated on the mesh. A first set of first points is generated for each element based on a request for modification, and a set of second points is generated for each element based on the set of first points and the request for modification. Then, the frames of the video stream can be transformed by modifying the elements of the object on the basis of the sets of first and second points and the mesh. In such method, a background of the modified object can be changed or distorted as well by tracking and modifying the background.

(74) In some examples, transformations changing some areas of an object using its elements can be performed by calculating characteristic points for each element of an object and generating a mesh based on the calculated characteristic points. Points are generated on the mesh, and then various areas based on the points are generated. The elements of the object are then tracked by aligning the area for each element with a position for each of the at least one element, and properties of the areas can be modified based on the request for modification, thus transforming the frames of the video stream. Depending on the specific request for modification properties of the mentioned areas can be transformed in different ways. Such modifications may involve changing color of areas; removing at least some part of areas from the frames of the video stream; including one or more new objects into areas which are based on a request for modification; and modifying or distorting the elements of an area or object. In various implementations, any combination of such modifications or other similar modifications may be used. For certain models to be animated, some characteristic points can be selected as control points to be used in determining the entire state-space of options for the model animation.

(75) In some examples of a computer animation model to transform image data using face detection, the face is detected on an image with use of a specific face detection algorithm (e.g., Viola-Jones). Then, an Active Shape Model (ASM) algorithm is applied to the face region of an image to detect facial feature reference points.

(76) Other methods and algorithms suitable for face detection can be used. For example, in some implementations, features are located using a landmark, which represents a distinguishable point present in most of the images under consideration. For facial landmarks, for example, the location of the left eye pupil may be used. If an initial landmark is not identifiable (e.g., if a person has an eyepatch), secondary landmarks may be used. Such landmark identification procedures may be used for any such objects. In some examples, a set of landmarks forms a shape. Shapes can be represented as vectors using the coordinates of the points in the shape. One shape is aligned to another with a similarity transform (allowing translation, scaling, and rotation) that minimizes the average Euclidean distance between shape points. The mean shape is the mean of the aligned training shapes.

(77) In various examples, a search for landmarks from the mean shape aligned to the position and size of the face determined by a global face detector is started. Such a search then repeats the steps of suggesting a tentative shape by adjusting the locations of shape points by template matching of the image texture around each point and then conforming the tentative shape to a global shape model until convergence occurs. In one or more systems, individual template matches are unreliable, and the shape model pools the results of the weak template matches to form a stronger overall classifier. The entire search is repeated at each level in an image pyramid, from coarse to fine resolution.

(78) A transformation system can capture an image or video stream on a client device (e.g., the client device **102**) and perform complex image manipulations locally on the client device **102** while maintaining a suitable user experience, computation time, and power consumption. The complex image manipulations may include size and shape changes, emotion transfers (e.g., changing a face from a frown to a smile), state transfers (e.g., aging a subject, reducing apparent age, changing gender), style transfers, graphical element application, and any other suitable image or video manipulation implemented by a convolutional neural network that has been configured to execute efficiently on the client device **102**.

(79) A computer animation model to transform image data can be used by a system where a user may capture an image or video stream of the user (e.g., a selfie) using a client device **102** having a neural network operating as part of a client application **104** operating on the client device **102**. The transformation system operating within the client application **104** determines the presence of a face within the image or video stream and provides modification icons associated with a computer animation model to transform image data, or the computer animation model can be present as associated with an interface described herein. The modification icons include changes that may be the basis for modifying the user's face within the image or video stream as part of the modification operation. Once a modification icon is selected, the transform system initiates a process to convert the image of the user to reflect the selected modification icon (e.g., generate a smiling face on the user). A modified image or video stream may be presented in a graphical user interface displayed on the client device **102** as soon as the image or video stream is captured, and a specified modification is selected. The transformation system may implement a complex convolutional neural network on a portion of the image or video stream to generate and apply the selected modification. That is, the user may capture the image or video stream and be presented with a modified result in real-time or near real-time once a modification icon has been selected. Further, the modification may be persistent while the video stream is being captured, and the selected modification icon remains toggled. Machine taught neural networks may be used to enable such modifications.

(80) The graphical user interface, presenting the modification performed by the transform system,

may supply the user with additional interaction options. Such options may be based on the interface used to initiate the content capture and selection of a particular computer animation model (e.g., initiation from a content creator user interface). In various implementations, a modification may be persistent after an initial selection of a modification icon. The user may toggle the modification on or off by tapping or otherwise selecting the face being modified by the transformation system and store it for later viewing or browse to other areas of the imaging application. Where multiple faces are modified by the transformation system, the user may toggle the modification on or off globally by tapping or selecting a single face modified and displayed within a graphical user interface. In some implementations, individual faces, among a group of multiple faces, may be individually modified, or such modifications may be individually toggled by tapping or selecting the individual face or a series of individual faces displayed within the graphical user interface.

(81) A story table **312** stores data regarding collections of messages and associated image, video, or audio data, which are compiled into a collection (e.g., a story or a gallery). The creation of a particular collection may be initiated by a particular user (e.g., each user for which a record is maintained in the entity table **304**). A user may create a “personal story” in the form of a collection of content that has been created and sent/broadcast by that user. To this end, the user interface of the client application **104** may include an icon that is user-selectable to enable a sending user to add specific content to his or her personal story.

(82) A collection may also constitute a “live story,” which is a collection of content from multiple users that is created manually, automatically, or using a combination of manual and automatic techniques. For example, a “live story” may constitute a curated stream of user-submitted content from various locations and events. Users whose client devices have location services enabled and are at a common location event at a particular time may, for example, be presented with an option, via a user interface of the client application **104**, to contribute content to a particular live story. The live story may be identified to the user by the client application **104**, based on his or her location. The end result is a “live story” told from a community perspective.

(83) A further type of content collection is known as a “location story,” which enables a user whose client device **102** is located within a specific geographic location (e.g., on a college or university campus) to contribute to a particular collection. In some examples, a contribution to a location story may require a second degree of authentication to verify that the end user belongs to a specific organization or other entity (e.g., is a student on the university campus).

(84) As mentioned above, the video table **314** stores video data that, in one example, is associated with messages for which records are maintained within the message table **302**. Similarly, the image table **316** stores image data associated with messages for which message data is stored in the entity table **304**. The entity table **304** may associate various augmentations from the augmentation table **310** with various images and videos stored in the image table **316** and the video table **314**.

(85) The database(s) **118** may also store an AR content usage table **318** that stores data indicating usage metrics for augmented reality content. In various examples, the AR content usage table **318** may include one or more data structures that store usage metrics for individual augmented reality content items. The metrics may indicate a number of interactions with augmented reality content by users of the client application **104**. In addition, the database(s) **118** may store an AR content users table **320** that indicates users of the client application **104** that interact with augmented reality content via the client application **104**. The AR content users table **320** may include one or more identifiers of users of the client application **104** that interact with one or more augmented reality content items. The one or more identifiers may be used to access profile information of the users of the client application **104** that interact with augmented reality content using the client application **104**. In one or more examples, the AR content users table **320** may store at least a portion of the profile information of users of the client application **104** that interact with augmented reality content via the client application **104**.

(86) FIG. 4 is a schematic diagram illustrating an example framework for content **400**, according to

some implementations. The content **400** may be generated by the client application **104**. In various examples, the content **400** may be generated by a first instance of the client application **104** and communicated to at least one of a second instance of the client application **104** or the server system **108**. In situations where the content **400** includes a message, the content **400** may be used to populate the message table **302** stored within the database(s) **118** and accessible by the application server **114**. In one or more implementations, the content **400** may be stored in memory as “in-transit” or “in-flight” data of at least one of client devices **102** or the application server **114**. The content **400** is shown to include at least a portion of the following components: content identifier **402**: a unique identifier that identifies the content **400**. content text payload **404**: text, to be generated by a user via a user interface of the client device **102**, and that is included in the content **400**. content image payload **406**: image data, captured by a camera component of a client device **102** or retrieved from a memory component of a client device **102**, and that is included in the content **400**. Image data for a sent or received content **400** may be stored in the image table **316**. content video payload **408**: video data, captured by a camera component or retrieved from a memory component of the client device **102**, and that is included in the content **400**. Video data for a sent or received content **400** may be stored in the video table **314**. content audio payload **410**: audio data, captured by a microphone or retrieved from a memory component of the client device **102**, and that is included in the content **400**. content augmentation data **412**: augmentation data (e.g., filters, stickers, or other annotations or enhancements) that represents augmentations to be applied to content image payload **406**, content video payload **408**, or content audio payload **410** of the content **400**. Augmentation data for a sent or received content **400** may be stored in the augmentation table **310**. content duration parameter **414**: parameter value indicating, in seconds, the amount of time for which one or more portions of the content **400** (e.g., the content image payload **406**, content video payload **408**, content audio payload **410**) are to be presented or made accessible to a user via the client application **104**. content geolocation parameter **416**: geolocation data (e.g., latitudinal and longitudinal coordinates) associated with the content payload of the message. Multiple content geolocation parameter **416** values may be included in the payload, each of these parameter values being associated with respect to content items included in the content (e.g., a specific image into within the content image payload **406**, or a specific video in the content video payload **408**). content story identifier **418**: identifier values identifying one or more content collections (e.g., “stories” identified in the story table **312**) with which a particular item in the content image payload **406** of the content **400** is associated. For example, multiple images within the content image payload **406** may each be associated with multiple content collections using identifier values. content tag **420**: each content **400** may be tagged with multiple tags, each of which is indicative of the subject matter of content included in the content payload. For example, where a particular image included in the content image payload **406** depicts an animal (e.g., a lion), a tag value may be included within the content tag **420** that is indicative of the relevant animal. Tag values may be generated manually, based on user input, or may be automatically generated using, for example, image recognition. content sender identifier **422**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the client device **102** on which the content **400** was generated and from which the content **400** was sent. content receiver identifier **424**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the client device **102** to which the content **400** is addressed. AR tracked features identifier(s) **426**: an identifier of features of augmented reality content corresponding to the content **400** that may have an amount of usage that is being monitored or tracked. In one or more examples, an AR tracked features identifier **426** may correspond to a feature of an augmented reality content item that is associated with a user interface element that is selectable by users of the client application **104**.

(87) The data (e.g., values) of the various components of content **400** may correspond to pointers to locations in tables within which the data is stored. For example, an image value in the content

image payload **406** may be a pointer to (or address of) a location within an image table **316**. Similarly, values within the content video payload **408** may point to data stored within a video table **314**, values stored within the content augmentation data **412** may point to data stored in an augmentation table **310**, values stored within the content story identifier **418** may point to data stored in a story table **312**, and values stored within the content sender identifier **422** and the content receiver identifier **424** may point to user records stored within an entity table **304**. Further, values of the AR tracked features identifier(s) **426** may point to data stored within a data structure that includes the AR content usage table **318**. In these scenarios, the data stored in the AR content usage table **318** for a respective AR tracked features identifier **426** may correspond to an amount of interaction with a feature of an augmented reality content item of the content **400** by users of the client application **104**.

(88) FIG. 5 is a diagrammatic representation illustrating an architecture **500** to analyze augmented reality content usage data, in accordance with one or more example implementations. The architecture **500** may include an AR content creator **502**. The AR content creator **502** may include one or more entities that create augmented reality content to be executed by the client application **104**. In one or more examples, the AR content creator **502** may include one or more users of the client application **104**. In additional examples, the AR content creator **502** may include one or more third parties that provide augmented reality content for the client application **104**. In additional examples, the AR content creator **502** may include a service provider that maintains and implements the server system **108**. In these scenarios, the AR content creator **502** may be a service provider that maintains and implements the client application **104**.

(89) In one or more illustrative examples, the AR content creator **502** may create the AR content item **136**. The AR content item **136** may be executed by the client application **104** to generate at least one of one or more visual effects or one or more audio effects with regard to user content accessible using the client application **104**. The AR content item **136** may include computer readable instructions that are executable by the client application **104** to modify at least one of the appearance or sound features of the AR content item **136**. For example, a user **504** of the client application may generate user content. The user content may include one or more content items that comprise at least one of image content, video content, or audio content. In various examples, the user content may be captured by the user **504** via one or more input devices of the client device **102**. To illustrate, the user **504** may utilize one or more cameras of the client device **102** to capture image content. In these situations, the AR content item **136** may modify the image content by modifying an appearance of one or more objects included in the user content. In additional examples, the user **504** may utilize one or more microphones of the client device **102** to capture audio content. In these scenarios, the AR content item **136** may modify at least a portion of the audio content included in the user content.

(90) The AR content creator **502** may provide AR content creation data **506** that is used to generate the AR content item **136**. The AR content creator **502** may provide the AR content creation data **506** to the server system **108**. In one or more implementations, the AR content creation system **128** may use the AR content creation data **506** to generate the AR content item **136**. In one or more examples, the AR content creation data **506** may be provided using one or more user interfaces. In various examples, the one or more user interfaces may be generated by the AR content creation system **128**. The one or more user interfaces may include one or more user interface elements to capture data that may be used to generate augmented reality content.

(91) In various examples, the AR content creation data **506** may indicate one or more interactions by users of the client application **104** that are to be monitored or tracked by the server system **108** with respect to the AR content item **136**. For example, the AR content creation data **506** may indicate that the number of times that the AR content item **136** is shared are to be tracked by the AR content usage system **130**. In additional examples, the AR content creation data **506** may indicate that the number of times that the AR content item **136** is included in a message are to be tracked by

the AR content usage system **130**. In further examples, the AR content creation data **506** may indicate that the number of times that the AR content item **136** is applied to user content is to be tracked by the AR content usage system **130**. The AR content creation data **506** may also indicate that usage of one or more features of the AR content item **136** is to be tracked. To illustrate, the AR content item **136** may include one or more features with which users of the client application may interact. In one or more examples, the AR content item **136** may include one or more features that may be at least one of selected, moved, or modified by users of the client application **104**. Interaction with a feature of the AR content item **136** may be based on user input obtained via one or more input devices of the client device **102**. For example, interaction with a feature of the AR content item **136** may include at least one of touching a user interface element of the AR content item **136** using a finger or other implement for at least a threshold period of time, hovering over a user interface element of the AR content item **136** using a finger or other implement, selecting and moving a user interface element of the AR content item **136**, swiping a user interface element of the AR content item **136**, performing a pinching operation (e.g., bringing two fingers closer together) with respect to a user interface element of the AR content item **136**, or an expanding operation (e.g., moving two fingers away from each other) with respect to a user interface element of the AR content item **136**.

(92) In one or more illustrative examples, the AR content item **136** may include an animation that includes a character with which the user **504** may interact. For example, the user **504** may cause the character to move one or more body parts. In these situations, the AR content creation data **506** may indicate that at least one of the body parts is a feature of the AR content item **136** for which interactions are to be tracked by the server system **108**. The user **504** may also cause the character to change locations within a user interface. In various examples, the AR content creation data **506** may indicate that modifying a location of the character may be an interaction with the AR content item **136** that is to be tracked by the server system **108**. Additionally, the user **504** may cause the character to make one or more audible noises. In these scenarios, the AR content creation data **506** may indicate that causing the character to produce at least one of the audible noises is a feature of the AR content item **136** that is to be tracked by the server system **108**.

(93) The AR content creation data **506** may be obtained via one or more templates that indicate at least one of features or interactions with respect to AR content items for which user interactions may be tracked by the server system **108**. In these scenarios, the AR content creator **502** may use a template to indicate at least one feature or interaction with respect to the AR content item **136** that is to be tracked by the server system **108**. Thus, in situations where one or more templates are used to indicate at least one of features or interactions of augmented reality content items that are to be tracked, the AR content creation system **128** implements a standardized approach to identifying at least one of features or interactions of augmented reality content items to be monitored or tracked by the server system **108**. In additional examples, the AR content creation system **128** may provide user interfaces that enable user driven selection of at least one of features or interactions of augmented reality content items that are to be tracked. To illustrate, the AR content creator **502** may use an input device to indicate any number of features of the AR content item **136**, interactions with the AR content item **136**, or both that are to be tracked by the server system **108**. In these situations, the AR content creation system **128** may implement a customized approach to identifying at least one of features or interactions of augmented reality content items that are to be monitored or tracked by the server system **108**. In one or more implementations, the AR content creation system **128** may implement a combination of a customized and standardized approach to the selection of at least one of features or interactions with AR content items that are to be monitored by the server system **108**.

(94) In the illustrative example of FIG. 5, the AR content item **136** may include a first AR interaction **508**, a second AR interaction **510**, up to an Nth AR interaction **512**. At least one of the first AR interaction, the second AR interaction **510**, or the Nth AR interaction **512** may be tracked

by AR content usage system **130**. For example, the AR content creator **502** may indicate in the AR content creation data **506** that the server system **108** is to monitor for at least one of the first AR interaction **508**, the second AR interaction **510**, or the Nth AR interaction **512**. In one or more examples, at least one of the first AR interaction **508**, the second AR interaction **510**, or the Nth AR interaction **512** may be related to features of the AR content item **136** with which the user **504** may interact. For example, at least one of the first AR interaction **508**, the second AR interaction **510**, or the Nth AR interaction **512** may correspond to an object or a portion of an object with which the user **504** may interact. In additional examples, at least one of the first AR interaction **508**, the second AR interaction **510**, or the Nth AR interaction **512** may correspond to additional actions that may be performed with respect to the AR content item **136**. To illustrate, at least one of the first AR interaction **508**, the second AR interaction **510**, or the Nth AR interaction **512** may correspond to the user **504** sharing user content related to the AR content item **136** with one or more users of the client application **104**. In various examples, user content that is related to the AR content item **136** may be shared by the user **504** via social networking functionality of the client application **104**. In additional examples, at least one of the first AR interaction **508**, the second AR interaction **510**, or the Nth AR interaction **512** may correspond to the user **504** generating a message that includes user content related to the AR content item **136**, where the message is accessible to recipients via the client application **104**. In one or more implementations, at least one of the first AR interaction **508**, the second AR interaction **510**, or the Nth AR interaction **512** may correspond to a user interface element that is displayed via the client application **104**.

(95) The first AR interaction **508** may correspond to a first usage metric **514**. The first usage metric **514** may indicate a frequency that users of the client application **104** perform the first AR interaction **508**. In one or more examples, the first usage metric **514** may indicate a number of times that one or more users of the client application **104** perform the first AR interaction **508**. In various examples, the first usage metric **514** may indicate a number of times that users of the client application **104** perform the first AR interaction **508** within a period of time, such as one hour, one day, one week, one month, etc. In additional examples, the first usage metric **514** may correspond to an average number of times that individual users of the client application **104** perform the first AR interaction **508**. The first usage metric **514** may be determined by the AR content usage system **130** based on data obtained from instances of the client application **104** that are executed by client devices **102** of users of the client application **104**.

(96) The first AR interaction **508** may also be associated with first user profile data **516**. The first user profile data **516** may include information about users of the client application **104** that performed the first AR interaction **508**. The first user profile data **516** may indicate at least one of demographic information or location information of users of the client application **104** that performed the first AR interaction **508**. The first user profile data **516** may also indicate content usage characteristics of users of the client application **104** that performed the first AR interaction **508**. The content usage characteristics may indicate sources of content consumed by users of the client application **104** that performed the first AR interaction **508**. The content usage characteristics may also indicate types of augmented reality content interacted with by users of the client application **104**. Further, the content usage characteristics may indicate individual AR content items that are interacted with by users of the client application **104**. In one or more examples, the content usage characteristics may indicate a frequency of interaction with at least one of one or more sources of content or types of content by users of the client application **104** that performed the first AR interaction **508**.

(97) In one or more examples, in response to obtaining information indicating that the first AR interaction **508** has been performed, the AR content usage system **130** may determine a user of the client application **104** that performed the first AR interaction **508**. In various examples, individual users of the client application **104** may be associated with an identifier. For example, the identifier may include a number of alphanumeric characters or symbols that uniquely identify an individual

user of the client application **104**. The identifier may be provided by an individual user of the client application **104** or the identifier may be determined by a service provider that implements and maintains at least one of the client application **104** or the server system **108**. In various examples, one or more actions performed by individual users of the client application **104** may generate data that includes an identifier of the user of the client application **104** and that indicates the one or more actions performed. The data may be sent to the server system **108** and analyzed by the AR content usage system **130** to determine the respective user of the client application **104** that performed the one or more actions. Additionally, the AR content usage system **130** may access profile information of the respective user that performed the one or more actions. To illustrate, the AR content usage system **130** may query one or more data stores, such as the database(s) **118**, using the identifier of the user of the client application **104** that performed the one or more actions to access profile information of the user stored by the one or more data stores. In this way, the AR content usage system **130** may associate one or more characteristics of the user that performed the one or more actions. The AR content usage system **130** may aggregate profile information of a number of users of the client application **104** that perform the first AR interaction **508** to determine a characterization of users of the client application **104** that perform the first AR interaction **508**.

(98) In a manner that is similar to the first AR interaction **508**, the second AR interaction **510** may be associated with a second usage metric **518** and second user profile data **520** and the Nth AR interaction **512** may be associated with an Nth usage metric **522** and Nth user profile data **524**. Additionally, the AR content usage system **130** may determine users of the client application **104**, such as the user **504**, that perform at least one of the second AR interaction **510** or the Nth AR interaction **512** based on identifiers of the users of the client application **104**.

(99) The content management system **132** may generate a dashboard **526** that is accessible to the AR content creator **502**. The dashboard **526** may include one or more user interfaces **528** that include information about the AR content item **136**. The information included in the dashboard **526** may include AR content usage metrics **530**. The AR content usage metrics **530** may be determined by the AR content usage system **130** and indicate quantitative measures indicating interactions by users of the client application **104** with respect to the AR content item **136**. In various examples, the AR content usage metrics **530** may indicate at least one of a frequency or a periodicity with which users of the client application **104** perform one or more of the first AR interaction **508**, the second AR interaction **510**, or the Nth AR interaction **512**. The AR content usage metrics **530** may be displayed via at least one of one or more charts, one or more tables, or one or more graphs.

(100) The dashboard **526** may also include AR content user characterization information **532**. The AR content user characterization information **532** may indicate characteristics of users of the client application **104** that perform at least one of the first AR interaction **508**, the second AR interaction **510**, or the Nth AR interaction **512**. The AR content user characterization information **532** may be generated by the AR content usage system **130**. In one or more examples, the AR content user characterization information **532** may include information about characteristics of users of the client application **104** that interact with the AR content item **136** in relation to other users of the client application **104**. For example, the AR content user characterization information **532** may indicate a number of users of the client application **104** that perform the first AR interaction **508** in relation to a number of users of the client application **104** that interact with other AR content items **136**. To illustrate, the AR content user characterization information **532** may indicate that users of the client application **104** that perform the first AR interaction **508** are more likely to interact with augmented reality content via the client application **104** than users of the client application **104** that do not interact with augmented reality content via the client application **104**. Additionally, the AR content user characterization information **532** may indicate at least one of one or more demographic features or one or more location features that correspond to at least a threshold number of users of the client application **104** that perform the first AR interaction **508**. In one or more illustrative examples, the AR content user characterization information **532** may indicate that

at least 50% of the users of the client application **104** that perform the first AR interaction **508** are between the ages of 25 and 40 and that at least 75% of the users of the client application **104** that perform the first AR interaction **508** live in a city with a population greater than 250,000 people. (101) Information included in the dashboard **526** for the AR content creator **502** may be based on dashboard requests **534** received by the server system **108** from one or more computing devices of the AR content creator **502**. The dashboard requests **534** may indicate one or more criteria with respect to at least one of content usage metrics or characteristics of users of the client application **104** that interacted with AR content generated by the AR content creator **502**. In one or more examples, the content management system **132** may provide one or more user interfaces that may be used by the AR content creator **502** to specify information to include in the dashboard **526**. The one or more user interfaces may include options indicating types of information to include in the dashboard **526**. In the illustrative example of FIG. 5, the server system **108** may send dashboard UI information **536** to the AR content creator **502** that includes information used to display the one or more user interfaces **528** of the dashboard **526**.

(102) In one or more examples, the content management system **132** may provide one or more user interfaces that include options that are selectable to include metrics indicating frequency of usage of the AR content item **136**, such as frequency of occurrence of at least one of the first AR interaction **508**, the second AR interaction **510**, or the Nth AR interaction **512**. The one or more user interfaces may also include options that are selectable to cause information to be displayed via the dashboard **526** that is related to one or more characteristics of users of the client application **104** that interact with the AR content item **136**. To illustrate, the content management system **132** may provide one or more user interfaces that include options selectable to view one or more demographic characteristics of users of the client application **104** that interact with the AR content item **136**, one or more location characteristics of users of the client application **104** that interact with the AR content item **136**, one or more content consumption history characteristics of users of the client application **104** that interact with the AR content item **136**, one or more combinations thereof, and so forth.

(103) Further, the content management system **132** may provide one or more user interfaces that include options that are selectable to indicate one or more visual formats used to present the information included in the dashboard **526**. For example, the content management system **132** may provide one or more user interfaces that include one or more options that are selectable to display at least one of the AR content usage metrics **530** or the AR content user characterization information **532** in a numerical format. In additional examples, the content management system **132** may provide one or more user interfaces that include one or more options that are selectable to display at least one of the AR content usage metrics **530** or the AR content user characterization information **532** in a tabular format. The content management system **132** may also provide one or more user interfaces that include one or more options that are selectable to display at least one of the AR content usage metrics **530** or the AR content user characterization information **532** in a graphical form, such as a pie graph, a line graph, a bar graph, combinations thereof, and the like.

(104) In one or more implementations, the content management system **132** may provide information indicating a target audience for the AR content item **136**. In various examples, the information related to the target audience for the AR content item **136** may be included in the dashboard UI information **536**. In additional examples, target audience information may be provided separate from the dashboard UI information **536**. In one or more examples, the target audience may be determined by the AR content usage system **130** based on usage metrics and user profile data of users of the client application **104** that interact with augmented reality content of the AR content creator **502**, such as the AR content item **136**. The target audience information may indicate at least one of demographic characteristics or location characteristics of users of the client application **104** that have at least a threshold likelihood of interacting with the AR content item **136**. For example, the target audience information may indicate at least one of demographic

characteristics or location characteristics of users of the client application **104** that have at least a threshold probability of performing the first AR interaction **508**. In addition, the target audience information may indicate at least one of demographic characteristics or locations characteristics of users of the client application **104** that have at least a threshold probability of performing the second AR interaction **510**. In various examples, the target audience information with respect to the first AR interaction **508** may be different from the target audience information with respect to the second AR interaction **510**. Further, the target audience information may indicate content consumption characteristics of users of the client application **104** that have at least a threshold probability of interacting with the AR content item **136**, such as performing the first AR interaction **508** or the second AR interaction **510**. The content consumption characteristics may include types of content (e.g., video content, audio content, image content, text content, augmented reality content, etc.) viewed or accessed by users of the client application **104**, sources of content viewed or accessed by users of the client application **104**, frequency of usage of one or more types of content via the client application **104**, frequency of usage of content from one or more sources via the client application **104**, one or more channels used to access content via the client application **104**, or one or more combinations thereof.

(105) After determining a target audience for the AR content item **136**, the server system **108** may obtain input from the AR content creator **502** indicating one or more channels that may be used to promote the AR content item **136** via the client application **104**. In one or more examples, the AR content creator **502** may indicate that the AR content item **136** is to be promoted to users of the client application **104** included in the target audience by presenting the AR content item **136** in a relatively high position of search results that are produced in response to search requests for AR content that are directed to one or more keywords associated with the AR content item **136**. In these scenarios, the AR content item **136** may be weighted in a manner that moves the AR content item **136** closer to the top of search results provided to users of the client application **104** included in the target audience that submit search requests to the server system **108**. In one or more additional examples, the AR content creator **502** may indicate that the AR content item **136** is to be promoted to users of the client application **104** included in the target audience by including the AR content item **136** in content recommendations provided via the client application **104**.

(106) In one or more implementations, a threshold probability of users of the client application **104** performing the first AR interaction **508** may be determined by analyzing usage metrics and user profile data for the first AR interaction **508** in relation to average usage metrics for at least a subset of additional AR content items accessible via the client application **104**. For example, the AR content usage system **130** may determine an average amount of usage of the AR content item **136** across a number of users of the client application **104** and then determine characteristics of users of the client application **104** that use the AR content item **136** more than the average amount of usage for the AR content item **136**. The AR content usage system **130** may determine a threshold probability for users of the client application **104** to be included in the target audience for the AR content item **136** based on the number of users of the client application **104** that may be included in the target audience. The AR content usage system **130** may determine a likelihood or probability of a user of the client application **104** being in the target audience for the AR content item **136** based on a measure of similarity between characteristics of the user and characteristics of users of the client application **104** that have at least a threshold amount of usage of the AR content item **136**. The likelihood or probability of the user of the client application **104** being in the target audience may be analyzed in relation to the threshold probability to determine whether or not to include the user in the target audience of the AR content item **136**.

(107) In one or more illustrative examples, the AR content item **136** may be related to one or more products, such as one or more beauty products. The one or more products may be available for purchase via the client application **104**. In various examples, the first AR interaction **508** may correspond to selection of a first product within a user interface generated by the client application

104, the second AR interaction **510** may correspond to selection of a second product within a user interface generated by the client application **104**, and the Nth AR interaction **512** may correspond to selection of an Nth product within a user interface generated by the client application **104**. In these scenarios, the first product may be associated with a first selectable user interface element, the second product may be associated with a second selectable user interface element, and the Nth product may be associated with an Nth selectable user interface element. In response to a user of the client application **104** performing the first AR interaction **508**, the second AR interaction **510**, or the Nth AR interaction **512**, one or more visual effects may be applied to user content. In one or more examples, the first AR interaction **508** may correspond to a first product and selection of a user interface element that corresponds to the first product may cause one or more visual effects related to characteristics of the first product to be applied to user content. In situations where the user content includes one or more facial features of a user of the client application **104** and the first product is a beauty product, selection of the first user interface element corresponding to the first product may cause the user content to be modified in a way that shows the first product being applied to the one or more facial features, such as applying eye shadow to a region around one or more eyes included in the user content. In one or more scenarios where the second AR interaction **510** corresponds to a second product, selection of the user interface element that corresponds to the second product may cause the user content to be modified in a way that shows the second product being applied to one or more facial features. The first product and the second product may be a same type of product (e.g., eye shadow), but may have different characteristics, such as different color, different tone, different brightness, different shading, different glossiness, and the like. In additional examples, the first product and the second product may be different types of products. To illustrate, the first product may be eye shadow and the second product may be lipstick. Further, the first product and the second product may be at least one of produced by different manufacturers or offered for purchase by different retailers. In other situations, the first product and the second product may be at least one of produced by the same manufacturer or offered for purchase by the same retailer.

(108) The information included in the dashboard **526** may also be related to one or more products offered for purchase via the client application **104**, where the products correspond to augmented reality content items generated by creators of augmented reality content. For example, the AR content creator **502** may be at least one of a manufacturer or a seller of one or more products and the AR content item **136** may include one or more user interface elements that correspond to one or more augmented reality content items that are related to one or more products manufactured or sold by the AR content creator **502**. In one or more examples, at least one of the first AR interaction **508**, the second AR interaction **510**, or the Nth AR interaction **512** may correspond to a product that is offered for purchase via the client application **104** by the AR content creator **502**. In one or more additional examples, the AR content item **136** may include user interface elements that correspond to products manufactured or sold by a number of different entities. To illustrate, the first AR interaction **508** may correspond to a first product that is offered for purchase by a first manufacturer or a first seller and the second AR interaction **510** may correspond to a second product that is offered for purchase by a second manufacturer or a second seller.

(109) In one or more illustrate examples, the dashboard **526** may include a number of options that are selectable to view information about one or more products related to augmented reality content items that are executed by the client application **104**. The dashboard **526** may arrange data related to the use of augmented reality content items that correspond to a number of products that are related to the augmented reality content items. The information included in the dashboard **526** may indicate a number of times or a frequency of use of augmented reality content items related to a number of products. In various examples, the information included in the dashboard **526** may indicate a number of times or a frequency that visual effects of augmented reality content items are applied to user content, where the visual effects are related to applying products to one or more

objects included in the user content.

(110) FIG. 6 illustrates a flowchart of one or more implementations of a process to analyze augmented reality content usage data. The processes may be embodied in computer-readable instructions for execution by one or more processors such that the operations of the processes may be performed in part or in whole by the functional components of at least one of the client application **104** or the server system **108**. Accordingly, the processes described below are by way of example with reference thereto, in some situations. However, in other implementations, at least some of the operations of the processes described with respect to FIG. 6 may be deployed on various other hardware configurations. The processes described with respect to FIG. 6 are therefore not intended to be limited to the server system **108** or client device **102** and can be implemented in whole, or in part, by one or more additional components. Although the described flowcharts can show operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed. A process may correspond to a method, a procedure, an algorithm, etc. The operations of methods may be performed in whole or in part, may be performed in conjunction with some or all of the operations in other methods, and may be performed by any number of different systems, such as the systems described herein, or any portion thereof, such as a processor included in any of the systems.

(111) FIG. 6 is a flowchart illustrating example operations of a process **600** to determine a characterization of client application users that interact with one or more augmented reality content items, according to one or more example implementations. The process **600** may include, at operation **602**, generating an augmented reality content item that is displayable via a client application. The client application may include the client application **104** and have at least one of social networking functionality or messaging functionality. The augmented reality content item may modify an appearance of user content, such as image content or video content captured by at least one camera of a client device. Modifying the appearance of the user content may include applying one or more visual effects with respect to one or more objects included in the user content. In scenarios where the augmented reality content item is related to a product, modifying the appearance of user content may include applying one more visual characteristics of the product to the user content. The augmented reality content item may include computer-readable instructions that are executable to modify user content via the client application.

(112) The augmented reality content item may be generated based on input obtained via one or more user interfaces. The one or more user interfaces may include one or more user interface elements to design augmented reality content items. For example, the one or more user interface elements may include one or more creative tools that produce one or more visual effects in response to execution of the augmented reality content item. The one or more user interfaces may include one or more user interface elements to upload content that may be used to generate the augmented reality content item.

(113) In addition, at operation **604**, the process **600** may include receiving user input indicating that usage of the augmented reality content item is to be tracked. In various examples, the user input may be captured via a standardized set of user interface elements that are selectable to indicate that at least one of one or more interactions or usage of one or more features of the augmented reality content item are to be tracked. The user input may be captured via a tool that may be used to build augmented reality content items and to indicate that usage of the augmented related content items is to be tracked. In one or more examples, the user interface elements that are selectable to indicate that at least one of one or more interactions or usage of one or more features of the augmented reality content item to be tracked are based on a type related to the augmented reality content item. For example, augmented reality content items that change the visual appearance of one or more objects, such as facial features, included in user content may include one or more first features that may be monitored and augmented reality content items that include an animation that is displayed

within user content may include one or more second features that may be monitored that are different from the one or more first features.

(114) The features of the augmented reality content item that may be tracked or monitored may include one or more portions of an object that a user of the client application may interact with. Additionally, the features of the augmented reality content item that may be tracked or monitored may be selected to apply visual effects related to a product to one or more objects included in user content. In one or more examples, features of the augmented reality content item that may be tracked or monitored may be selected to initiate a financial transaction to purchase at least one product offered for purchase via the client application.

(115) Further, the user input may indicate one or more types of interaction with the augmented reality content item that are to be tracked or monitored. To illustrate, the user input may indicate that sharing of the augmented reality content item by users of the client application is to be tracked. In addition, the user input may indicate that messages related to the augmented reality content item are to be tracked. In one or more implementations, adding the augmented reality content to a group of augmented reality content items activated by users of the client application may also be tracked. In various examples, the group of reality content items may be associated with an account of a user of the client application and may be displayed to the user, such as via a carousel user interface element. The user input may also indicate that a number of times that the augmented reality content item is applied to user content is to be tracked. In one or more examples, the augmented reality content item may be applied to user content in response to selection of a user interface element that corresponds to the augmented reality content item and that causes changes to user content that may include at least one of video content, image content, text content, or audio content.

(116) At operation **606**, the process **600** may include determining an amount of interaction with the augmented reality content item by one or more users of the client application. In one or more implementations, a server system may monitor data received from instances of the client application executed by respective client devices of users of the client application for information indicating that an interaction has occurred with respect to the augmented reality content item. For example, the client application may be executable to generate data in response to interactions with augmented reality content items and send the data to the server system. In various examples, data related to interaction with the augmented reality content item may be generated based on interaction with at least one user interface element that corresponds to the augmented reality content item. In response to receiving data from the client application indicating an interaction with an augmented reality content item, the server system may store information related to the interaction in a data structure corresponding to the augmented reality content item. In additional scenarios, individual instances of the client application may collect, store, and analyze interactions with augmented reality content items.

(117) The amount of interaction with the augmented reality content item may include a number of times that the augmented reality content item has been at least one of applied to user content, shared with other users of the client application, or included in a message. The amount of interaction with the augmented reality content item may also include a number of times that one or more features of the augmented reality content item have been interacted with by users of the client application. In one or more examples, the amount of interaction with the augmented reality content item may correspond to a frequency with which users of the client application interact with the augmented reality content item.

(118) Further, the process **600** may include, at operation **608**, determining profile information of the one or more users of the client application that interacted with the augmented reality content item. In one or more examples, the client application may determine an identifier of individual users of the client application that interacted with the augmented reality content item. The client application may also send the respective identifiers of the users of the client application that interact with the augmented reality content item to the server system. The server system may use

the user identifiers to parse one or more data stores to determine information about the users of the client application that interacted with the augmented reality content item. The profile information obtained from the one or more data stores may include at least one of demographic information, location information, or content consumption information. The profile information may be based on one or more requests from the creator of the augmented reality content item to identify specified characteristics of users of the client application that interacted with the augmented reality content item. The profile information may also be based on characteristics of users of the client application that may be predictive of users of the client application interacting with one or more augmented reality content items or interacting with one or more types of augmented reality content items.

(119) At operation **610**, the process **600** may include determining a characterization of users of the augmented reality content item based on the profile information. The characterization may indicate characteristics of users of the client application that interacted with the augmented reality content item. In one or more examples, the characterization may indicate characteristics that are common among users of the client application that have at least a threshold amount of interaction with the augmented reality content item. For example, a server system may determine an amount of interaction with the augmented reality content item that corresponds to a subset of users of the client application. To illustrate, the server system may determine users of the client application that have an amount of interaction with the augmented reality content item that falls within the top 25% of users of the client application that interacted with the augmented reality content item. The server system may then determine characteristics that the users within the subset have in common. In additional examples, the server system may determine a range of values for one or more characteristics of users included in the subset. In various examples, the server system may determine characteristics that are common across a percentage of users of the client application that have at least a threshold amount of interaction with the augmented reality content item.

(120) In one or more implementations, the characterization of users of the client application that interact with the augmented reality content item may be provided to a creator of the augmented reality content item. For example, one or more user interfaces may be provided to the creator of the augmented reality content item that indicate characteristics of users of the client application that interact with the augmented reality content item. In one or more examples, the characterization of users of the client application that interact with the augmented reality content item may be used to determine a target audience for the augmented reality content item. The target audience may include at least one of users of the client application that have not interacted with the augmented reality content item or users of the client application that have not added the augmented reality content item to a collection of augmented reality content items of the respective users.

(121) In various examples, information about the augmented reality content item may be provided to at least a portion of the users of the client application included in the target audience. To illustrate, information about the augmented reality content item may be provided in a content discovery section of the client application for at least a portion of the users of the client application included in the target audience. In additional examples, information about the augmented reality content item may be given a higher priority in search results related to search requests provided by at least a portion of the users of the client application in the target audience. In at least some examples, the search requests may include one or more keywords that correspond to the augmented reality content item. In one or more examples, members of the target audience may receive recommendations for augmented reality content that include the augmented reality content item. Also, members of the target audience may receive advertisements as part of a campaign to bring awareness about the augmented reality content item.

(122) FIG. 7 is an illustration of a user interface **700** that includes features that may be incorporated into augmented reality content, according to one or more example implementations. The user interface **700** may be displayed via a display device of the client device **102** and produced by a client application executed by the client device **102**, such as the client application **104**. The user

interface **700** may include a view of one or more products **702** that may be used to modify an appearance of one or more objects included in user content. In various examples, the one or more products **702** may include one or more beauty products. In the illustrative example of FIG. 7, the one or more products may include different colors of eye shadow that may be applied to a region proximate to eyes of individuals. In one or more examples, the different colors of eye shadow may be at least one of manufactured or sold by one or more entities. That is, in some cases, the colors of eye shadow may be manufactured or sold by a same entity, such as a same manufacturer or retailer, while in other scenarios, at least some of the different colors of eye shadow may be manufactured or sold by different entities, such as different manufacturers or retailers.

(123) The one or more products **702** may be included in a field of view of at least one camera of the client device **102**, such as a camera **704**. In one or more implementations, image content of the one or more products **702** may be captured by the camera **704** and used to create augmented reality content. For example, image content that corresponds to the one or more products may be used to generate an augmented reality content item that is executed to produce one or more visual effects that show the one or more products being applied to one or more objects included in user content. In various examples, the augmented reality content item may apply one or more characteristics of the one or more products to the user content. The user content may include additional image content or video content captured by the camera **704**.

(124) The user interface **700** may also include a group of user interface elements **706** that correspond to a collection of augmented reality content items that may be applied to user content captured via the camera **704**. Each of the user interface elements **706** may be selectable to execute a respective augmented reality content item with respect to user content. Executing an augmented reality content item may cause one or more visual effects to be applied to user content in order to change an appearance of one or more objects included in the user content. In the illustrative example of FIG. 7, the group of user interface elements **706** may include a user interface element **708** that corresponds to an augmented reality content item that may be executed to apply one or more beauty products to an individual included in user content captured by the camera **704**. In various examples, the user interface **700** may be a first user interface in a series of user interfaces that are generated in relation to the augmented reality content item that corresponds to the user interface element **708**.

(125) FIG. 8 is an illustration of a user interface **800** that includes user content **802** to which an augmented reality content item may be applied, according to one or more example implementations. The user interface **800** may be displayed via a display device of the client device **102** and produced by a client application executed by the client device **102**, such as the client application **104**. In one or more examples, the user interface **800** may be generated after the user interface **700** and may include an additional user interface in a series of user interfaces that corresponds to an augmented reality content item related to the user interface element **708**. In various examples, the user content **802** may be captured by the camera **704** and may include one or more body parts of an individual. In the illustrative example of FIG. 8, the user content **802** may include a forearm of an individual.

(126) FIG. 9 is an illustration of a user interface **900** that includes augmented reality content that is applied to user content with a number of features of the augmented reality content being associated with a selectable user interface element, according to one or more example implementations. The user interface **900** may be displayed via a display device of the client device **102** and produced by a client application executed by the client device **102**, such as the client application **104**. The user interface **900** may show modified user content **902** that may be a modified version of the user content **802** of FIG. 8. In various examples, the modified user content **902** may be the result of the one or more products shown in the user interface **700** of FIG. 7 being applied to the user content shown in the user interface **800**. In one or more examples, the modified user content **902** may include different colors of eye shadow applied to the forearm of an individual. In one or more

examples, the user interface **900** may be generated after the user interface **800** of FIG. **8** and may include a further user interface in a series of user interfaces associated with the augmented reality content item related to the user interface element **708**.

(127) In one or more implementations, each eye shadow color shown in FIG. **9** may correspond to a selectable user interface element. For example, Color 1 may be associated with a first user interface element **904** and Color 2 may be associated with a second user interface element **906**. In various examples, the first user interface element **904** may be selectable to execute a first augmented reality content item that applies a first eye shadow color to a region proximate to one or more eyes of an individual and the second user interface element may be selectable to execute a second augmented reality content item that applies a second eye shadow color to a region proximate to one or more eyes of an individual. In one or more additional examples, the first user interface element **904** may be selectable to display information related to a first product associated with the first eye shadow color and the second user interface element may be selectable to display information related to a second product associated with the second eye shadow color.

(128) FIG. **10** is a representation of an environment **1000** that includes a client device **102** that displays a user interface **1002** including information about one or more augmented reality content items of an AR content creator, according to one or more example implementations. The user interface **1002** may be displayed via a display device of the client device **102** and produced by a client application executed by the client device **102**, such as the client application **104**. In one or more examples, the client device **102** may be operated by an AR content creator or a representative of an AR content creator. In one or more illustrative examples, the user interface **1002** may be displayed in relation to the dashboard **526** of FIG. **5**.

(129) The user interface **1002** may include a first section **1004** that includes an identifier **1006** of an AR content creator. The first section **1004** may also include a user interface element **1008** that enables the selection of one or more options to display information about augmented reality content items created by the AR content creator. The user interface **1002** may also include a second section **1010** that includes information about a respective augmented reality content item of a number of augmented reality content items generated by the AR content creator. The second section **1010** may include an identifier, such as a name, of the augmented reality content item, a date and time that the augmented reality content item was updated, and a status of the augmented reality content item.

(130) In addition, the user interface **1002** may include a third section **1012** that includes a number of usage metrics with respect to the augmented reality content item. For example, the third section **1012** may indicate a number of times that users of the client application **104** have viewed the augmented reality content item. Further, the third section **1012** may indicate a number of times that users of the client application **104** have played or otherwise executed the augmented reality content item and a number of times that users of the client application **104** have shared the augmented reality content item. The third section **1012** may also include additional information about the augmented reality content item, such as a system identifier for the augmented reality content item. The system identifier may be an identifier used by the server system **108** to track the usage of the augmented reality content item. Further, the third section **1012** may include a user interface element **1014** that is selectable to share information about the augmented reality content item.

(131) FIG. **11** is a representation of an environment **1100** that includes a client device **102** that displays an additional user interface **1102** including information about one or more augmented reality content items of an AR content creator, according to one or more example implementations. The user interface **1102** may be displayed via a display device of the client device **102** and produced by a client application executed by the client device **102**, such as the client application **104**. In one or more examples, the client device **102** may be operated by an AR content creator or a representative of an AR content creator. In one or more illustrative examples, the user interface **1102** may be displayed in relation to the dashboard **526** of FIG. **5**.

(132) The user interface **1102** may include a first section **1104** that includes a number of options

that are selectable to view information related to one or more services provided by a service provider that collects and analyzes usage information about augmented reality content items. In various examples, the service provide may at least one of implement or maintain the server system **108**. The first section **1104** may also include a number of options that are selectable to view information related to one or more reports associated with a respective service option that provide data corresponding to usage of augmented reality content items. In the illustrative example of FIG. **11**, the information shown in the user interface **1102** corresponds to augmented reality content items that are related to a number of products. In various examples, the augmented reality content corresponding to the products may be executed to apply characteristics of the respective products to one or more objects included in user content by modifying a visual appearance of the one or more objects based on visual effects related to the respective products. Additionally, with respect to the illustrative example of FIG. **11**, a user interface element **1106** corresponding to Report Option 3 has been selected, where Report Option 3 is related to Service Option 2.

(133) The user interface **1102** may also include a second section **1108** that includes information corresponding to a selected option included in the first section **1104**. For example, the second section **1108** may include usage information for augmented reality content items related to a number of products included in a product category. In addition to indicating the product category and the product name, the second section **1108** may indicate an identifier of individuals products, such as a numerical identifier related to an individual product, a number of uses of an augmented reality content item corresponding to an individual product, and a number of times that user content was captured and modified by an augmented reality content item. Further, the second section **1108** may indicate a brand corresponding to individual products, where the brand may indicate at least one of a seller or a manufacturer of a respective product.

(134) FIG. **12** is a representation of an environment **1200** that includes a client device **102** that displays a user interface **1202** including information that characterizes users of a client application that interact with augmented reality content items according to one or more example implementations. The user interface **1202** may be displayed via a display device of the client device **102** and produced by a client application executed by the client device **102**, such as the client application **104**. In one or more examples, the client device **102** may be operated by an AR content creator or a representative of an AR content creator. In one or more illustrative examples, the user interface **1202** may be displayed in relation to the dashboard **526** of FIG. **5**.

(135) The user interface **1202** may include a number of options **1204** that are selectable to generate different views of information that indicate characteristics of users of the client application **104** that use one or more augmented reality content items to modify user content via the client application **104**. The user interface **1202** may also include a user interface element **1206** that is operable to select a date range for information related to usage of one or more augmented reality content items. In addition, the user interface **1202** may include a first section **1208** that includes information related to usage metrics for one or more augmented reality content items, such as daily unique views of one or more augmented reality content items, total unique views of one or more augmented reality content items, daily core users of one or more augmented reality content items, and an average amount of time that users of the client application **104** viewed the one or more augmented reality content items. In various examples, a core user of an augmented reality content item may indicate a user of the client application **104** that uses an augmented reality content item at least a threshold number of times or for a threshold amount of time within a given period of time.

(136) Further, the user interface **1202** may include a second section **1210** that indicates gender demographics over time related to the use of one or more augmented reality content items. The gender demographics may indicate a number of users of the client application **104** that use one or more augmented reality content items during a period of time and the respective gender of those users. Additionally, the user interface **1202** may include a third section **1212** that indicate age demographics over time related to the use of one or more augmented reality content items. The age

demographics may indicate a number of users of the client application **104** that use one or more augmented reality content items during a period of time and the respective age of those users.

(137) FIG. **13** is a block diagram illustrating components of a machine **1300**, according to some example implementations, able to read instructions from a machine-readable medium (e.g., a machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **13** shows a diagrammatic representation of the machine **1300** in the example form of a computer system, within which instructions **1302** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1300** to perform any one or more of the methodologies discussed herein may be executed. As such, the instructions **1302** may be used to implement modules or components described herein. The instructions **1302** transform the general, non-programmed machine **1300** into a particular machine **1300** programmed to carry out the described and illustrated functions in the manner described. In alternative implementations, the machine **1300** operates as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1300** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1300** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smart phone, a mobile device, a wearable device (e.g., a smart watch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1302**, sequentially or otherwise, that specify actions to be taken by machine **1300**. Further, while only a single machine **1300** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions **1302** to perform any one or more of the methodologies discussed herein.

(138) The machine **1300** may include processors **1304**, memory/storage **1306**, and I/O components **1308**, which may be configured to communicate with each other such as via a bus **1310**. In an example implementation, the processors **1304** (e.g., a central processing unit (CPU), a reduced instruction set computing (RISC) processor, a complex instruction set computing (CISC) processor, a graphics processing unit (GPU), a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a radio-frequency integrated circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, a processor **1312** and a processor **1314** that may execute the instructions **1302**. The term “processor” is intended to include multi-core processors **1304** that may comprise two or more independent processors (sometimes referred to as “cores”) that may execute instructions **1302** contemporaneously. Although FIG. **10** shows multiple processors **1304**, the machine **1300** may include a single processor **1312** with a single core, a single processor **1312** with multiple cores (e.g., a multi-core processor), multiple processors **1312**, **1314** with a single core, multiple processors **1312**, **1314** with multiple cores, or any combination thereof.

(139) The memory/storage **1306** may include memory, such as a main memory **1316**, or other memory storage, and a storage unit **1318**, both accessible to the processors **1304** such as via the bus **1310**. The storage unit **1318** and main memory **1316** store the instructions **1302** embodying any one or more of the methodologies or functions described herein. The instructions **1302** may also reside, completely or partially, within the main memory **1316**, within the storage unit **1318**, within at least one of the processors **1304** (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **1300**. Accordingly, the main memory **1316**, the storage unit **1318**, and the memory of processors **1304** are examples of machine-readable media.

(140) The I/O components **1308** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **1308** that are included in a particular

machine **1300** will depend on the type of machine. For example, portable machines such as mobile phones will likely include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **1308** may include many other components that are not shown in FIG. **10**. The I/O components **1308** are grouped according to functionality merely for simplifying the following discussion and the grouping is in no way limiting. In various example implementations, the I/O components **1308** may include user output components **1320** and user input components **1322**. The user output components **1320** may include visual components (e.g., a display such as a plasma display panel (PDP), a light emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The user input components **1322** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location or force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

(141) In further example implementations, the I/O components **1308** may include biometric components **1324**, motion components **1326**, environmental components **1328**, or position components **1330** among a wide array of other components. For example, the biometric components **1324** may include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components **1326** may include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The environmental components **1328** may include, for example, illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometer that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment. The position components **1330** may include location sensor components (e.g., a GPS receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

(142) Communication may be implemented using a wide variety of technologies. The I/O components **1308** may include communication components **1332** operable to couple the machine **1300** to a network **1334** or devices **1336**. For example, the communication components **1332** may include a network interface component or other suitable device to interface with the network **1334**. In further examples, communication components **1332** may include wired communication components, wireless communication components, cellular communication components, near field communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **1336** may be another machine **1300** or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

(143) Moreover, the communication components **1332** may detect identifiers or include components operable to detect identifiers. For example, the communication components **1332** may

include radio frequency identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **1332**, such as location via Internet Protocol (IP) geo-location, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

(144) FIG. **14** is a block diagram illustrating system **1400** that includes an example software architecture **1402**, which may be used in conjunction with various hardware architectures herein described. FIG. **14** is a non-limiting example of a software architecture and it will be appreciated that many other architectures may be implemented to facilitate the functionality described herein. The software architecture **1402** may execute on hardware such as machine **1300** of FIG. **10** that includes, among other things, processors **1304**, memory/storage **1306**, and input/output (I/O) components **1308**. A representative hardware layer **1404** is illustrated and can represent, for example, the machine **1300** of FIG. **13**. The representative hardware layer **1404** includes a processing unit **1406** having associated executable instructions **1408**. Executable instructions **1408** represent the executable instructions of the software architecture **1402**, including implementation of the methods, components, and so forth described herein. The hardware layer **1404** also includes at least one of memory or storage modules memory/storage **1410**, which also have executable instructions **1408**. The hardware layer **1404** may also comprise other hardware **1512**.

(145) In the example architecture of FIG. **14**, the software architecture **1402** may be conceptualized as a stack of layers where each layer provides particular functionality. For example, the software architecture **1402** may include layers such as an operating system **1414**, libraries **1416**, frameworks/middleware **1418**, applications **1420**, and a presentation layer **1422**. Operationally, the applications **1420** or other components within the layers may invoke API calls **1424** through the software stack and receive messages **1426** in response to the API calls **1424**. The layers illustrated are representative in nature and not all software architectures have all layers. For example, some mobile or special purpose operating systems may not provide a frameworks/middleware **1418**, while others may provide such a layer. Other software architectures may include additional or different layers.

(146) The operating system **1414** may manage hardware resources and provide common services. The operating system **1414** may include, for example, a kernel **1428**, services **1430**, and drivers **1432**. The kernel **1428** may act as an abstraction layer between the hardware and the other software layers. For example, the kernel **1428** may be responsible for memory management, processor management (e.g., scheduling), component management, networking, security settings, and so on. The services **1430** may provide other common services for the other software layers. The drivers **1432** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **1432** include display drivers, camera drivers, Bluetooth® drivers, flash memory drivers, serial communication drivers (e.g., Universal Serial Bus (USB) drivers), Wi-Fi® drivers, audio drivers, power management drivers, and so forth depending on the hardware configuration.

(147) The libraries **1416** provide a common infrastructure that is used by at least one of the applications **1420**, other components, or layers. The libraries **1416** provide functionality that allows other software components to perform tasks in an easier fashion than to interface directly with the underlying operating system **1414** functionality (e.g., kernel **1428**, services **1430**, drivers **1432**). The libraries **1416** may include system libraries **1434** (e.g., C standard library) that may provide functions such as memory allocation functions, string manipulation functions, mathematical functions, and the like. In addition, the libraries **1416** may include API libraries **1436** such as media libraries (e.g., libraries to support presentation and manipulation of various media format such as

MPEG4, H.264, MP3, AAC, AMR, JPG, PNG), graphics libraries (e.g., an OpenGL framework that may be used to render two-dimensional and three-dimensional in a graphic content on a display), database libraries (e.g., SQLite that may provide various relational database functions), web libraries (e.g., WebKit that may provide web browsing functionality), and the like. The libraries **1416** may also include a wide variety of other libraries **1438** to provide many other APIs to the applications **1420** and other software components/modules.

(148) The frameworks/middleware **1418** (also sometimes referred to as middleware) provide a higher-level common infrastructure that may be used by the applications **1420** or other software components/modules. For example, the frameworks/middleware **1418** may provide various graphical user interface functions, high-level resource management, high-level location services, and so forth. The frameworks/middleware **1418** may provide a broad spectrum of other APIs that may be utilized by the applications **1420** or other software components/modules, some of which may be specific to a particular operating system **1414** or platform.

(149) The applications **1420** include built-in applications **1440** and third-party applications **1442**. Examples of representative built-in applications **1440** may include, but are not limited to, a contacts application, a browser application, a book reader application, a location application, a media application, a messaging application, or a game application. Third-party applications **1442** may include an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform, and may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or other mobile operating systems. The third-party applications **1442** may invoke the API calls **1424** provided by the mobile operating system (such as operating system **1414**) to facilitate functionality described herein.

(150) The applications **1420** may use built-in operating system functions (e.g., kernel **1428**, services **1430**, drivers **1432**), libraries **1416**, and frameworks/middleware **1418** to create UIs to interact with users of the system. Alternatively, or additionally, in some systems, interactions with a user may occur through a presentation layer, such as presentation layer **1422**. In these systems, the application/component “logic” can be separated from the aspects of the application/component that interact with a user.

Glossary

(151) “CARRIER SIGNAL,” in this context, refers to any intangible medium that is capable of storing, encoding, or carrying transitory or non-transitory instructions **1302** for execution by the machine **1300**, and includes digital or analog communications signals or other intangible medium to facilitate communication of such instructions **1302**. Instructions **1302** may be transmitted or received over the network **110**, **1334** using a transitory or non-transitory transmission medium via a network interface device and using any one of a number of well-known transfer protocols.

(152) “CLIENT DEVICE,” in this context, refers to any machine **1300** that interfaces to a communications network **110**, **1334** to obtain resources from one or more server systems or other client devices **102**. A client device **102** may be, but is not limited to, a mobile phone, desktop computer, laptop, PDAs, smart phones, tablets, ultra books, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network **110**, **1334**.

(153) “COMMUNICATIONS NETWORK,” in this context, refers to one or more portions of a network **110**, **1334** that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network **110**, **1334** or a portion of a network may include a wireless or cellular network and the coupling

may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other type of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1×RTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard setting organizations, other long range protocols, or other data transfer technology.

(154) “EPHEMERAL MESSAGE,” in this context, refers to a message that is accessible for a time-limited duration. An ephemeral message may be a text, an image, a video, and the like. The access time for the ephemeral message may be set by the message sender. Alternatively, the access time may be a default setting, or a setting specified by the recipient. Regardless of the setting technique, the message is transitory.

(155) “MACHINE-READABLE MEDIUM,” in this context, refers to a component, device, or other tangible media able to store instructions **1302** and data temporarily or permanently and may include, but is not limited to, random-access memory (RAM), read-only memory (ROM), buffer memory, flash memory, optical media, magnetic media, cache memory, other types of storage (e.g., erasable programmable read-only memory (EEPROM)) and/or any suitable combination thereof. The term “machine-readable medium” may be taken to include a single medium or multiple media (e.g., a centralized or distributed database, or associated caches and servers) able to store instructions **1302**. The term “machine-readable medium” shall also be taken to include any medium, or combination of multiple media, that is capable of storing instructions **1302** (e.g., code) for execution by a machine **1300**, such that the instructions **1302**, when executed by one or more processors **1304** of the machine **1300**, cause the machine **1300** to perform any one or more of the methodologies described herein. Accordingly, a “machine-readable medium” refers to a single storage apparatus or device, as well as “cloud-based” storage systems or storage networks that include multiple storage apparatus or devices. The term “machine-readable medium” excludes signals per se.

(156) “COMPONENT,” in this context, refers to a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, APIs, or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various example implementations, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein.

(157) A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an ASIC. A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may

include software executed by a general-purpose processor **1304** or other programmable processor. Once configured by such software, hardware components become specific machines (or specific components of a machine **1300**) uniquely tailored to perform the configured functions and are no longer general-purpose processors **1304**. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software) may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering implementations in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor **1304** configured by software to become a special-purpose processor, the general-purpose processor **1304** may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor **1312**, **1314** or processors **1304**, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time.

(158) Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In implementations in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output.

(159) Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information). The various operations of example methods described herein may be performed, at least partially, by one or more processors **1304** that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors **1304** may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors **1304**. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor **1312**, **1314** or processors **1304** being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors **1304** or processor-implemented components. Moreover, the one or more processors **1304** may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines **1300** including processors **1304**), with these operations being accessible via a network **110** (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine **1300**, but deployed across a number of machines. In some example implementations, the processors **1304** or processor-implemented components may be located in a

single geographic location (e.g., within a home environment, an office environment, or a server farm). In other example implementations, the processors **1304** or processor-implemented components may be distributed across a number of geographic locations.

(160) “PROCESSOR,” in this context, refers to any circuit or virtual circuit (a physical circuit emulated by logic executing on an actual processor **1304**) that manipulates data values according to control signals (e.g., “commands,” “op codes,” “machine code,” etc.) and which produces corresponding output signals that are applied to operate a machine **1300**. A processor **1304** may, for example, be a CPU, a RISC processor, a CISC processor, a GPU, a DSP, an ASIC, a RFIC or any combination thereof. A processor **1304** may further be a multi-core processor having two or more independent processors **1304** (sometimes referred to as “cores”) that may execute instructions **1302** contemporaneously.

(161) “TIMESTAMP,” in this context, refers to a sequence of characters or encoded information identifying when a certain event occurred, for example giving date and time of day, sometimes accurate to a small fraction of a second.

(162) Changes and modifications may be made to the disclosed implementations without departing from the scope of the present disclosure. These and other changes or modifications are intended to be included within the scope of the present disclosure, as expressed in the following claims.

Claims

1. A method comprising: capturing, by one or more computing devices, augmented reality content creation data that corresponds to generating one or more augmented reality content items that are executable to modify user content captured by one or more cameras; causing, by the one or more computing devices, a first user interface of an augmented reality content creator dashboard to be displayed that includes first values of one or more characteristics of users that interacted with the one or more augmented reality content items; and causing, by the one or more computing devices, a second user interface of the augmented reality content creator dashboard to be displayed that indicates second values of one or more usage metrics indicating interactions of the users with the one or more augmented reality content items.
2. The method of claim 1, wherein; the one or more augmented reality content items correspond to one or more products; and the one or more usage metrics are related to purchasing the one or more products.
3. The method of claim 1, wherein the interactions of the users with the one or more augmented reality content items correspond to a number of times that an individual augmented reality content item of the one or more augmented reality content items has been shared by one or more of the users.
4. The method of claim 1, wherein the one or more characteristics of the users that interacted with the one or more augmented reality content items include one or more locations of the users that interacted with the one or more augmented reality content items.
5. The method of claim 1, wherein the interactions of the users with the one or more augmented reality content items correspond to a number of times that an individual augmented reality content item of the one or more augmented reality content items has been viewed by one or more of the users.
6. The method of claim 1, wherein the interactions of the users with the one or more augmented reality content items correspond to a number of messages related to an augmented reality content item of the one or more augmented reality content items.
7. The method of claim 1, wherein the one or more usage metrics indicate a frequency of interaction with an augmented reality content item of the one or more augmented reality content items.
8. The method of claim 1, wherein at least one of the first user interface or the second user interface

include a group of user interface elements with individual user interface elements of the group of user interface elements corresponding to individual augmented reality content items of a plurality of augmented reality content items of an augmented reality content creator.

9. The method of claim 1, wherein the second user interface includes a user interface element that is selectable to view additional values of the one or more usage metrics with respect to an additional augmented reality content item of a plurality of augmented reality content items of an augmented reality content creator.

10. The method of claim 1, wherein the second user interface indicates at least one of a total number of views of a plurality of augmented reality content items of an augmented reality content creator, a total number of shares of the plurality of augmented reality content items of the augmented reality content creator, or a total number of messages related to the plurality of augmented reality content items of an augmented reality content creator.

11. The method of claim 1, wherein that one or more augmented reality content items correspond to one or more advertisements.

12. The method of claim 1, wherein at least one of the first values of one or more characteristics of the users that interacted with the one or more augmented reality content items or the second values of the one or more usage metrics are displayed via at least one of one or more charts, one or more tables, or one or more graphs.

13. A computing apparatus comprising: one or more hardware processors; and memory storing computer-readable instructions that, when executed by the one or more hardware processors, cause the one or more hardware processors to perform operations comprising: capturing augmented reality content creation data that corresponds to generating one or more augmented reality content items that are executable to modify user content captured by one or more cameras; causing a first user interface of an augmented reality content creator dashboard to be displayed that includes first values of one or more characteristics of users that interacted with the one or more augmented reality content items; and causing a second user interface of the augmented reality content creator dashboard to be displayed that indicates second values of one or more usage metrics indicating interactions of the users with the one or more augmented reality content items.

14. The computing apparatus of claim 13, wherein the one or more augmented reality content items are related to an advertisement campaign.

15. The computing apparatus of claim 13, wherein capturing the augmented reality content creation data includes capturing, via one or more user interface elements, information indicating at least one of a user characteristic or a usage metric that is to be tracked in relation to the one or more augmented reality content items.

16. The computing apparatus of claim 13, wherein the memory stores additional computer-readable instructions that, when executed by the one or more hardware processors, cause the one or more hardware processors to perform additional operations comprising: displaying one or more additional user interfaces to capture input indicating one or more channels to be used to promote the one or more augmented reality content items, the one or more channels including at least one of presenting the one or more augmented reality content items in results for searches of augmented reality content that correspond or providing the one or more augmented reality content items in recommendations for augmented reality content to one or more users.

17. The computing apparatus of claim 13, wherein: the first user interface includes one or more first options that are individually selectable to display at least a portion of the first values corresponding to a characteristic of the one or more characteristics of the users that interacted with the one or more augmented reality content items; and the second user interface includes one or more second options that are individually selectable to display at least a portion of the second values corresponding to a usage metric of the one or more usage metrics.

18. One or more computer-readable storage media storing computer-readable instructions that, when executed by one or more hardware processors, cause the one or more hardware processors to

perform operations comprising: capturing augmented reality content creation data that corresponds to generating one or more augmented reality content items that are executable to modify user content captured by one or more cameras; causing a first user interface of an augmented reality content creator dashboard to be displayed that includes first values of one or more characteristics of users that interacted with the one or more augmented reality content items; and causing a second user interface of the augmented reality content creator dashboard to be displayed that indicates second values of one or more usage metrics indicating interactions of the users with the one or more augmented reality content items.

19. The one or more computer-readable storage media of claim 18, wherein: an augmented reality content item of the one or more augmented reality content items includes a feature of a product that corresponds to a user interface element displayed in conjunction with the augmented reality content item, the user interface element being selectable to apply one or more visual effects to one or more user content items; and the one or more usage metrics indicate a number of selections of the user interface element.

20. The one or more computer-readable storage media of claim 18, storing additional computer-readable instructions that, when executed by one or more hardware processors, cause the one or more hardware processors to perform additional operations comprising: causing one or more additional user interfaces to be displayed as part of a tool to build augmented reality content items, the one or more additional user interfaces including one or more user interface elements that enable selection of one or more features of an augmented reality content item, wherein selection of the one or more features of the augmented reality content item using the tool causes interaction with the one or more features to be tracked in relation to the one or more usage metrics and the one or more features are included in a template of the tool.
